Organizational Conditions for AI-Driven Firm Performance

A Systematic Review of Empirical Evidence, 2015–2026

Bachelor's Thesis
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Abstract

Artificial intelligence (AI) is available to every firm, and a technology every rival can buy is a weak basis for advantage. Whether and when AI investments pay off is an empirical question. This thesis asks under what conditions firm-level AI investments translate into competitive advantage and superior firm performance. It answers the question with a systematic literature review: a documented Scopus search over the publication window 2015 to 2026 identified 432 records, which a journal-quality filter and two screening stages reduced to 67 empirical studies, each coded by two independent coders against a fixed scheme. Across the 67 studies, the evidence leans positive, with 38 reporting positive average effects, and it is conditional almost throughout: 63 studies identify at least one condition that decides whether the investment reaches its outcome. The review orders these conditions into six groups: complementary investments, organizational capabilities, leadership and governance, firm characteristics, environmental conditions, and the dose, credibility, and timing of the investment itself. Competitive advantage, kept analytically separate from performance throughout, is measured in only 15 studies and almost never as persistence against imitation. The findings support theories that locate AI's payoff in what surrounds the technology rather than in the technology itself, and they mark persistence as the field's largest open question.

Keywords: artificial intelligence; AI investment; firm performance; competitive advantage; systematic literature review; organizational conditions

Contents

Chapter 1

Introduction

Whether investing in artificial intelligence (AI) makes a firm more competitive is, in Kemp's words, a theoretical puzzle (Kemp, 2024, p. 618). The promise is large: intelligent algorithms are expected to automate a substantial share of work and to open new products and product-development processes. The same literature lists strategic obstacles: AI can be myopic and hard for managers to steer, and, built on explicit knowledge, it resembles a general-purpose technology (Kemp, 2024, p. 618). A technology every competitor can buy is a poor candidate for an advantage. What firms get out of buying it is an empirical question.

The empirical evidence has grown quickly and does not agree with itself. On German panel data, AI use raises productivity for the yes/no measure and the intensity measure alike (Czarnitzki et al., 2023, pp. 189, 201). Growth from AI investments concentrates among the larger firms, and industry concentration rises alongside (Babina et al., 2024, p. 1). Investors, on average, react to AI investment announcements with a drop of 1.77 percent on the announcement day, and firms with weak IT bases or poor credit ratings lose more (Lui et al., 2022, p. 373). And where adoption is observed over time, it destroys old capabilities while creating new ones at the human-machine boundary (Krakowski et al., 2023, p. 1425). Positive, negative, and conditional findings stand side by side, often for the same outcome. The disagreement itself points at the question worth asking: under which circumstances do the investments pay off.

That is the question this thesis asks: *under what conditions do firm-level AI investments translate into competitive advantage and superior firm performance?* The two outcome constructs are kept analytically separate throughout, because a firm can improve its performance without holding anything a rival cannot copy; the theoretical grounds for that separation are laid out in Chapter 2.

The thesis answers the question with a systematic literature review of the empirical evidence, searched over the publication window 2015 to 2026. A documented Scopus query identified 432 records, which a journal-quality filter, title-abstract screening, and full-text screening reduced to a final corpus of 67 empirical studies; every study was coded along a fixed scheme by two independent coders, and the resulting coding table is the review's core deliverable. Chapter 2 condenses the four theoretical lenses used to interpret the evidence: general-purpose technologies, the resource-based view, automation and augmentation, and situated AI. Chapter 3 documents the search, screening, and coding protocol. Chapter 4 presents the findings in two parts, a descriptive mapping of the evidence base and a synthesis of the conditions the studies identify, ordered into six groups. Chapter 5 interprets the findings against the theoretical lenses and names the limits of what the reviewed literature can show, and Chapter 6 concludes.

Chapter 2

Theoretical Background

2.1 Competitive advantage and firm performance

Competitive advantage is measured in 15 of the 67 studies reviewed here, and the remaining 52 report performance outcomes only. Barney separates the two constructs by what rivals are able to do: a firm holds a competitive advantage while it runs a value-creating strategy no rival, actual or potential, is running at the same time, and that advantage counts as sustained only once those rivals also cannot reproduce what the strategy yields (Barney, 1991, p. 102). Sustained in that sense says nothing about elapsed time. It asks whether the advantage is still there after rivals have given up trying to copy it (Barney, 1991, p. 102). This review therefore keeps the two apart throughout, because a firm can raise its margins without holding anything a rival is unable to copy.

Where both constructs appear, the separation is visible in the survey items. Chatterjee et al. ask about rivals directly: one of their competitive-advantage items has respondents affirm that successful implementation of AI-based customer relationship management "will help an organization to win over its competitor which has not yet implemented AI-CRM for B2B relationship management" (Chatterjee et al., 2021, p. 215). Their performance items stay inside the firm and ask whether the same system improves "operational efficiency" and speeds up decisions (Chatterjee et al., 2021, p. 215).

Mehta et al. give competitive advantage a position between adoption and results: they test it as the channel through which AI adoption reaches performance, and predict a positive effect on performance in its own right (Mehta et al., 2025, p. 550).

Not every study that names competitive advantage builds its own instrument. Hossain et al. measure sustained competitive advantage with items carried over from earlier work (Hossain et al., 2022, p. 247). The construct label alone says little about what a study measured; the items decide.

2.2 Artificial intelligence as a general-purpose technology

Economists file AI under general-purpose technologies (GPTs), and that classification already predicts an uneven payoff. A GPT earns the label because it makes "new and complementary production methods" available, which can raise output over time (Czarnitzki et al., 2023, p. 188). Those complements are the expensive part. Before output moves, a firm has to redesign how work is organised, retrain staff, rework its software and accumulate managerial experience, and because none of that accumulation is booked as an asset, measured productivity dips before it climbs (Brynjolfsson et al., 2021, pp. 333–334). The lag is years, not quarters.

Two studies in this corpus show how differently that plays out at firm level. On German innovation-survey panel data, Czarnitzki et al. measure a clear productivity gain from AI use, for the yes/no measure and for the intensity measure alike (Czarnitzki et al., 2023, pp. 189, 201). Fontanelli et al. ask a different question of French firms and find that AI use raises the *volatility* of productivity growth; the effect sits with firms that buy AI from outside providers, and it weakens where more of the payroll is made up of ICT engineers and technicians (Fontanelli et al., 2025, pp. 1–2). Across the two results, the reading this review takes forward is that the same AI investment can stand behind either outcome; what differs between the firms is the complements.

2.3 The resource-based view and the imitability problem

Across the studies reviewed here, the resource-based view (RBV) is the recurring reason given for expecting AI to pay off. Barney sets four conditions on a resource before it can hold an advantage in place: it has to be valuable, rare among competitors and imperfectly imitable, and no substitute may do the same strategic job (Barney, 1991, pp. 105–106). Alwakid and Dahri restate the four conditions as the VRIN acronym (Alwakid & Dahri, 2025, p. 14). That is a shorthand Barney himself comes close to when he lists "rareness, inimitability, non-substitutability" among the characteristics that can qualify a firm attribute as a source of advantage, adding that the attribute becomes a resource only once it exploits an opportunity or neutralizes a threat (Barney, 1991, p. 106). Hossain et al. put the same argument in terms of what a firm does with what it owns, and tie the long-run case to holdings rivals cannot match (Hossain et al., 2022, pp. 241–242).

Purchased AI fails that test, and the RBV literature says so plainly. Krakowski et al. note that AI costs almost nothing to reproduce and that little stands in the way of copying it, so where AI substitutes a human capability, the RBV predicts that the advantage resting on that capability will erode (Krakowski et al., 2023, p. 1426). Their own evidence relocates the advantage rather than denying it: what resists imitation is the combination a firm builds, not either side of it (Krakowski et al., 2023, p. 1446).

Two consequences follow for this review. The RBV speaks to competitive advantage and not to performance, so a study reporting higher margins has not thereby shown that anything is defensible against rivals; the separation of the two constructs in Section 2.1 is a theoretical requirement, not only a coding convention. And if the resource itself is imitable, the variation worth studying lies in what surrounds it, which is why the conditions under which AI pays off are treated here as the object of analysis rather than as controls.

2.4 Automation and augmentation

The same investment can be deployed in two ways, and the strategy literature keeps them apart. Raisch and Krakowski define automation as machines taking a task over from a human, and augmentation as humans and machines working on a task together (Raisch & Krakowski, 2021, p. 192). Their argument against the popular advice to prefer augmentation is that the two cannot be held apart in practice. For any single task at any single moment the firm picks one approach, and that choice makes the two contradictory; widen the frame to several tasks over several years and each turns out to produce the other, which makes them interdependent (Raisch & Krakowski, 2021, pp. 195–197).

Corpus studies pick the distinction up as a design choice with different returns. Zebec and Indihar Štemberger treat the two as the use cases through which AI adoption reaches performance, automation paying off in efficiency and augmentation in what people and machines manage jointly (Zebec & Indihar Štemberger, 2024, p. 7). Their model then returns no significant direct effect of process automation on process performance once tasks become complex, and the recommendation they draw from it is a boundary condition: automate what is routine, augment what is ambiguous (Zebec & Indihar Štemberger, 2024, pp. 16–17). Where the line falls is itself unstable. In the fragrance industry the machine took over part of the perfumers' problem solving and could not touch the part that depends on smelling a scent and judging how people will react to it (Krakowski et al., 2023, p. 1445).

2.5 Situated AI

Kemp starts from the puzzle the two previous sections produce: AI is available to everyone, so buying it should not yield an advantage, yet firms plainly differ in what they get out of it. His answer is situated AI, which he defines as AI hemmed in by the firm's own experience, structure and relationships (Kemp, 2024, p. 618). Three properties of the technology stand in the way, and the first of them is the imitability problem from Section 2.3 under another name: AI is generic, explicit and myopic. Being generic puts AI on a level with general human capital, which competitors can acquire and put to work much as the adopter does (Kemp, 2024, p. 620).

What a firm does control is the situating. Kemp names three activities: grounding, which picks the experiences a firm's own AI learns from; bounding, which works on the experiences feeding a competitor's AI; and recasting, which keeps reworking the algorithms and the routines around them so the AI stays aligned with the rest of the firm's activities (Kemp, 2024, pp. 618–619). The vocabulary is conditional throughout. None of the three is a property of the technology, and each is something a firm may or may not do.

Empirical work in the corpus starts from the same reading. Shi et al. take the mixed results of earlier work as a sign that what AI yields depends on intangible complements, on governance, and on managerial capability bound to its context, and they test that claim on Chinese listed firms, with data assets as the channel and managerial capability as the moderator (Shi et al., 2025, p. 1). Kemp's paper is conceptual, so what it offers are propositions rather than findings. They are the propositions this review is positioned to examine, because the conditions Kemp derives are the conditions the coded studies report.

Chapter 3

Method

3.1 Review design

This thesis answers its research question with a systematic literature review. The choice follows from the question itself: whether firm-level AI investments pay off, and under what conditions, is an empirical matter on which evidence has accumulated quickly and across scattered outlets. Management scholars pick the method for the transparency and comprehensiveness of its process (Rojon et al., 2021, p. 196). Every step that shapes the final corpus is documented here so that each decision can be retraced.

Scopus, accessed under the WU license, is the single data source. All studies that enter the review come from one documented query. Sources outside this corpus appear only as theory anchors in Chapter 2 and as method literature in this chapter; the review findings rest on the corpus alone. Scopus does not index grey literature or preprints, so the review is limited to peer-reviewed journal articles by design.

3.2 Search strategy

The search was run on 17 July 2026 through the Scopus Search API with the following query:

```
TITLE(("artificial intelligence" OR "machine learning" OR "deep
learning" OR "AI")) AND TITLE-ABS-KEY(("invest" OR "invests" OR
"invested" OR "investing" OR "investment" OR "investments" OR
"investor" OR "investors" OR adopt* OR implement*) AND ("competitive
advantage" OR "firm performance" OR "financial performance" OR "firm
productivity" OR "firm value" OR "market value" OR "firm growth")) AND
SUBJAREA(BUSI OR ECON OR DECI) AND PUBYEAR > 2014 AND PUBYEAR < 2027
AND DOCTYPE(ar) AND LANGUAGE(english) AND SRCTYPE(j)
```

The query places the AI vocabulary in the title, which keeps the search on studies where AI is the subject rather than a side topic, and the investment and outcome vocabulary in title, abstract, and keywords. The outcome list names the constructs the corpus uses, including firm value and firm growth. Decision sciences is included as a subject area because several operations research journals that publish firm-level AI evidence are indexed there, not under business or economics. The investment terms are written out instead of using a wildcard: `invest*` also matches *investigate* and its inflections, which inflated an earlier version of the query by 75 records that studied AI without treating it as an investment.

Three studies were fixed in advance as retrieval benchmarks (Babina et al., 2024; Krakowski et al., 2023; Lui et al., 2022): a sound query must find all of them. The final query does, and it returned 432 unique records.

3.3 Selection criteria

Two layers of criteria decide what stays in the corpus. The first is journal quality: a record remains when its journal is listed in the AJG 2024 with a rating of at least 2. Most records this rule removes appeared in journals not listed in the AJG at all. One fixed edition of the ranking applies to every record regardless of publication year, which keeps the rule uniform and non-selective, at the price of rating older records against a newer list.

The second layer holds four content criteria. A study stays in when it (1) is empirical, (2) works at the firm level, (3) treats firms' AI investment or adoption as its object of study, and (4) measures a performance or competitive-advantage outcome. Case studies count as empirical. Studies that survey managers stay in when the outcome is measured at the firm level.

Exclusions carry one of five codes, assigned in a fixed order in which the first matching code wins, so that every excluded record has exactly one reason and the counts are unambiguous: **E1** not empirical (conceptual work, reviews, simulations without firm data); **E2** not at the firm level (individual, team, or country); **E3** AI is the study's analysis tool rather than the firms' investment object; **E4** no performance or competitive-advantage outcome (for example adoption intention or user satisfaction); **E5** wrong publication type. At the title and abstract stage the rule in cases of doubt was to keep the record and let the full text decide.

3.4 Screening process

Figure 3.1 shows the full chain. The journal-quality filter reduced the 432 records to 174; this step required no reading, since the AJG rating attaches to the journal and was matched by ISSN. The 174 remaining records were screened on title and abstract against the four content criteria, which removed 93 (E1: 20, E2: 7, E3: 39, E4: 27; E5 was never needed). The full texts of the remaining 81 records were then assessed for eligibility: each full text was checked against the same four criteria, a check for which the method and results sections of an article are usually decisive. This stage excluded another fourteen: twelve on content grounds and two whose full texts could not be obtained through the university license or open access. The review proceeds with 67 studies. Screening was AI-assisted in the same way as the coding described below; every decision was confirmed by the author.

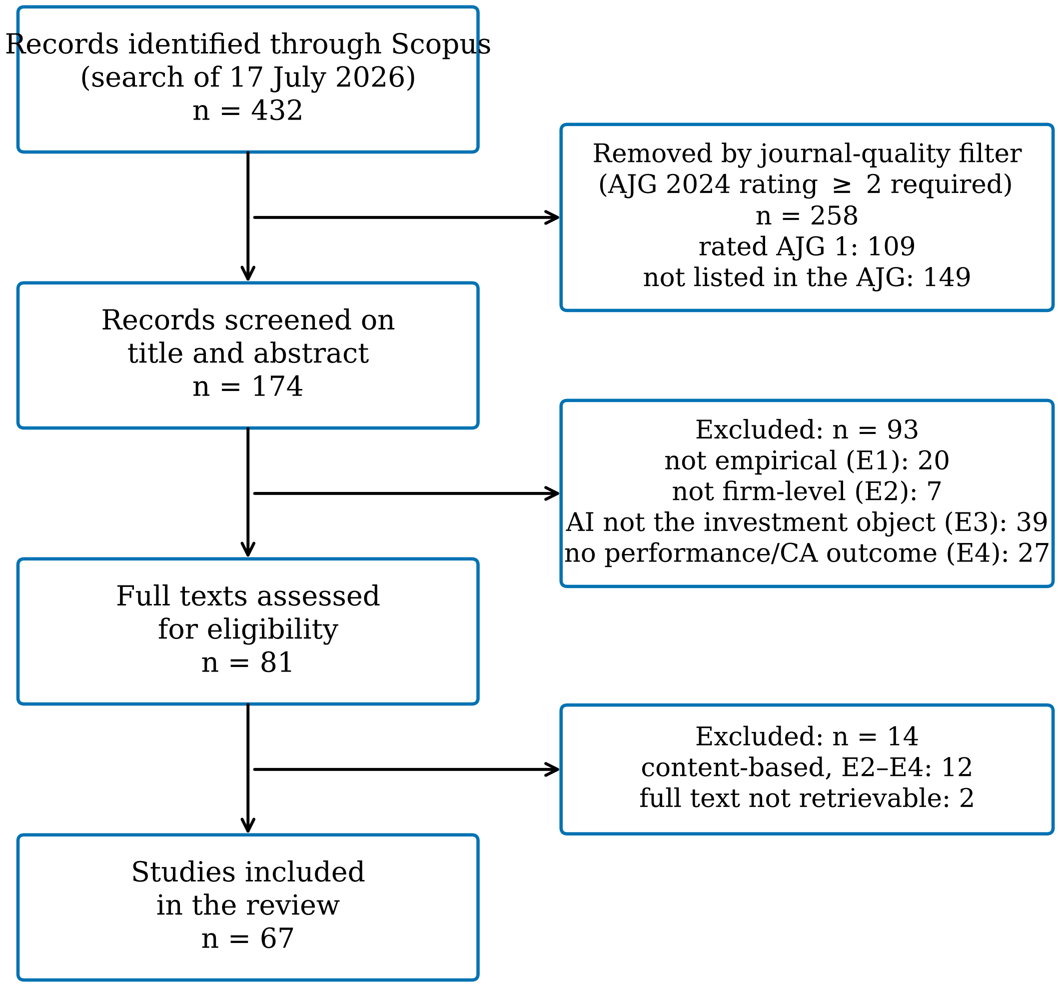

Figure 3.1: PRISMA-style flow of identification, screening, and inclusion. Source: own representation.

3.5 Coding

Each included study occupies one row of the coding table, split into study characteristics and findings (Appendix, Tables A.1 and A.2). The scheme was fixed before coding began; its fields operationalize the research question rather than following a single published template. The study-characteristics side records how the evidence was produced. It holds the theoretical lens a study works with, its method, its sample and period, the country and industry covered, and the way it measures AI investment, for example as a survey construct, through AI-related resumes and job postings, through AI patents, or as adoption announcements. The findings side records what the evidence says. It holds the outcome construct (performance, competitive advantage, or both), the concrete performance and competitive-advantage measures, the direction of the effect, the identified conditions with their roles as moderators, mediators, complements, or thresholds, a one-sentence key finding, and quality notes, for example when an outcome rests on managers' self-reports.

The direction of the effect takes one of five values. Positive, negative, and null describe the sign of the average effect, or its absence; a study coded positive can still identify conditions that strengthen or weaken it. Mixed means the finding splits across outcome measures, as when revenue rises but costs do not fall. Conditional means the sign or size of the effect depends on identified conditions; for the research question these studies carry the most weight. Two rules keep the cells close to the studies. A cell contains only what a study explicitly reports, so a sample period or industry the paper never states stays empty instead of being inferred. And when a study's value fits none of the predefined categories, the cell carries the study's own wording; nothing is forced into the nearest category.

The outcome side separates superior firm performance from competitive advantage. A study is coded as measuring competitive advantage only when it measures a distinct construct, such as a validated competitive-advantage scale or genuinely competitor-comparative evidence; a performance gain alone is coded as performance. In the first case, competitive advantage appears as a construct of its own, defined against other organizations in the same market and tested as a separate hypothesis (Chatterjee et al., 2021, p. 209). In the second, firms investing in AI grow faster in sales and employment, a performance finding that claims no position relative to competitors, and the study is coded as performance (Babina et al., 2024, p. 1). Each construct has its own measurement column; the table therefore also shows how rarely competitive advantage is measured at all.

Every study was coded twice in independent runs, the second run blind to the first; both runs were AI-assisted. Agreement between the runs was 90% for method, 88% for the outcome construct, and 73% for effect direction, the last after a documented pass that harmonized the first run's early direction codes to the codified rule. All 29 studies on which the runs disagreed on at least one coded field were adjudicated by the author against the full texts, with the decision rules recorded in writing, and a fixed-seed sample of ten agreement rows was re-checked the same way without finding a joint error. For studies never flagged during screening, the coding read doubled as the full-text eligibility check; an eligibility failure at this stage would have been documented back into the screening records.

The conditions column also provides the structure of the synthesis in Section 4.2: every condition entry was assigned to one of six thematic groups, and a study stands in several groups when it identifies several conditions. The grouping followed the same dual principle as the coding. Judgment cases went blind to a second, independent AI run, and the author decided every one of them.

Chapter 4

Results

This chapter reports the evidence in two steps. Section 4.1 maps the 67 included studies descriptively: where the evidence comes from and how it is produced. Section 4.2 then synthesizes the conditions under which firm-level AI investments translate into performance gains or competitive advantage. All counts, groupings, and figures in this chapter are the author's own compilation from the coding table; the full table is reported in the appendix.

4.1 Descriptive mapping of the evidence base

The search window ran from 2015 to 2026, yet the earliest included study dates from 2020 (Figure 4.1). Growth is concentrated at the end of that window: 44 of the 67 studies, two thirds of the corpus, appeared in 2025 and 2026 alone. The evidence base is young.

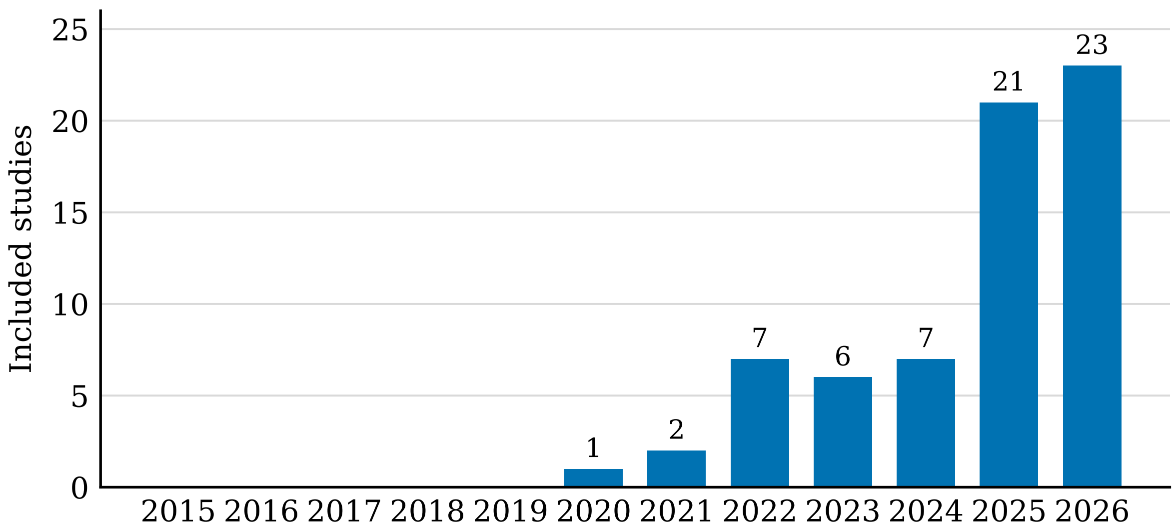

Figure 4.1: Included studies per publication year, search window 2015–2026 (n = 67). Source: own representation based on the coding table.

The 67 studies are spread across 45 journals, with no dominant outlet: Industrial Marketing Management and the Journal of Business Research lead with five studies each, followed by Technology in Society, the International Review of Financial Analysis, and the International Journal of Information Management with four each. Journal quality within the corpus is skewed toward the middle of the AJG scale (Figure 4.2): 27 studies come from journals rated 2, another 33 from journals rated 3, and seven from journals rated 4 or 4*.

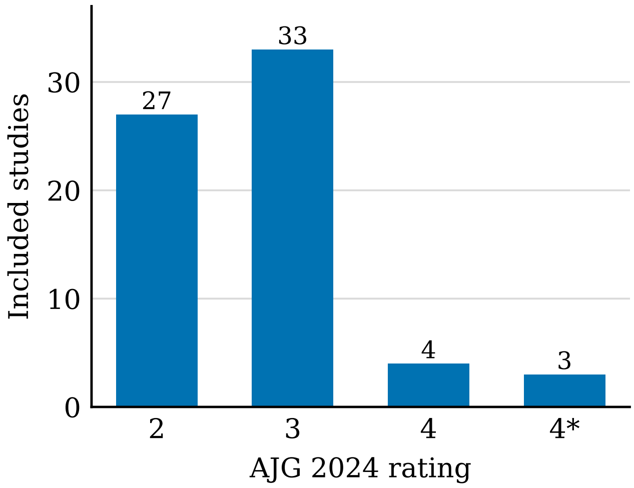

Figure 4.2: Included studies by AJG 2024 journal rating (n = 67). Source: own representation based on the coding table.

Two research designs dominate the corpus (Figure 4.3): panel econometrics on archival firm data (27 studies) and survey-based structural equation modeling (24 studies). The remaining sixteen studies combine several methods (six), examine stock market reactions in event studies (four), or build on case evidence (three); fsQCA, data envelopment analysis, and survey-based regression appear once each. The way studies operationalize AI investment splits the corpus in half: 33 studies rely on managers' self-reports through survey constructs, 31 on archival measures, and three on qualitative observation. Disclosure text mining (seven studies), patent-based proxies (six), workforce-based measures such as AI job postings (five), and investment announcements (four) account for most of the archival measures; the remaining nine rest on single-study instruments, for example procurement records or exposure to an AI policy shock.

Study samples cluster in two countries (Figure 4.4): the USA and China account for 16 studies each, together almost half of the corpus. India follows with seven studies, and another seven draw on multi-country samples. European firms appear in ten studies: four multi-country European samples plus six single-country ones. Most samples span several industries: 39 studies are cross-industry, ten focus on manufacturing, and the rest examine single settings such as healthcare or high-tech ventures.

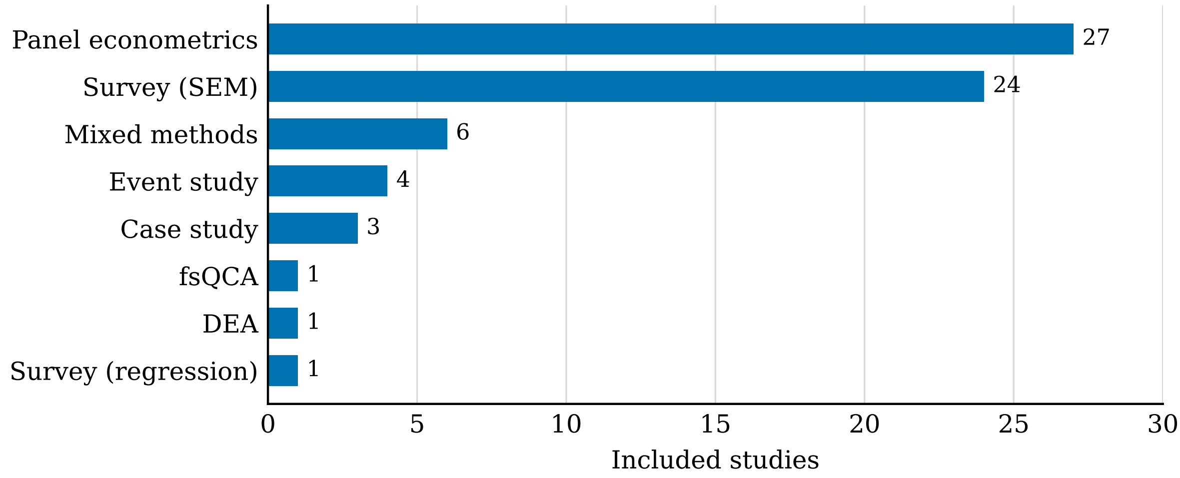

Figure 4.3: Included studies by research design (n = 67). Source: own representation based on the coding table.

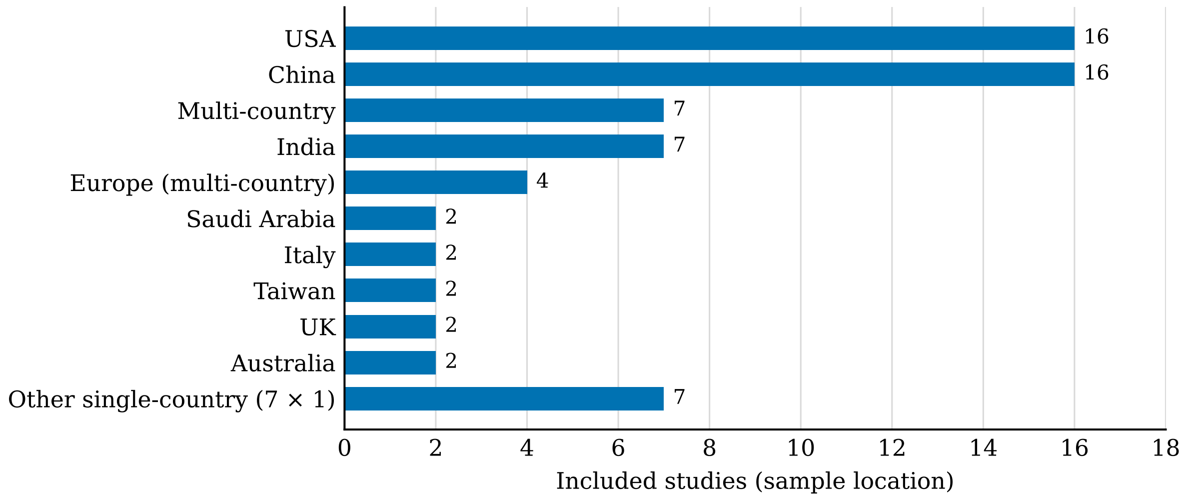

Figure 4.4: Included studies by sample location (n = 67). Source: own representation based on the coding table.

Following the distinction between superior firm performance and competitive advantage introduced in Chapter 2, the coding recorded which of the two constructs each study measures (Figure 4.5). The result is one-sided: 52 of the 67 studies measure firm performance only. Six measure competitive advantage as a distinct construct, and nine measure both. Measured competitive advantage is thus rare in this literature: 15 studies, under a quarter of the corpus. Effect directions spread across four of the five codes, and no study is coded null: 38 studies report a positive average effect of AI investment on their outcome, 20 an effect that exists only through a mechanism or under specific circumstances, five results that split across outcome indicators, and four negative effects. Among the 15 studies with a competitive-advantage construct, ten find positive and five conditional effects; none reports a negative one.

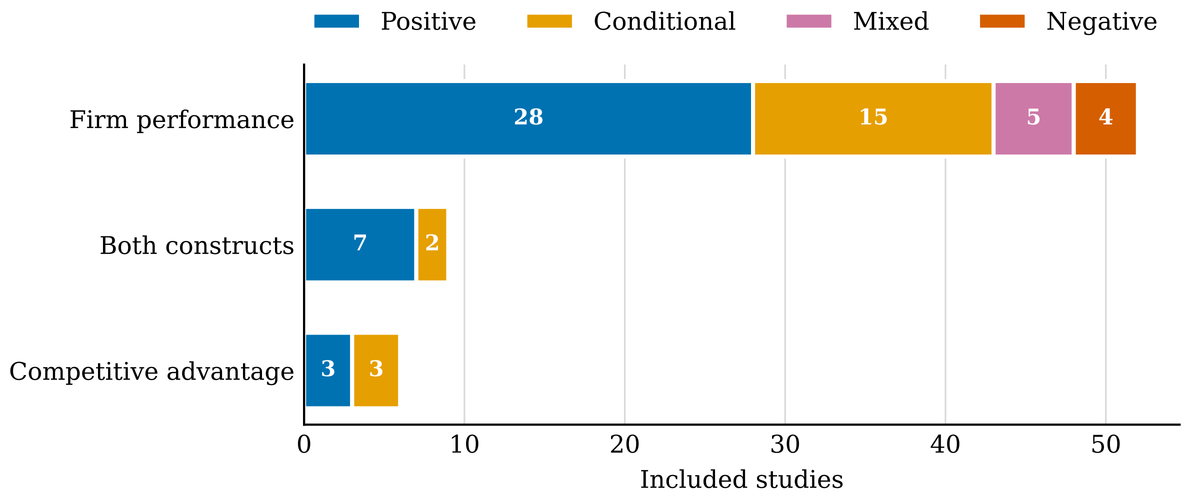

Figure 4.5: Outcome construct by reported effect direction (n = 67). Source: own representation based on the coding table.

Of the 67 studies, 63 identify at least one moderator, mediator, complement, or threshold that shapes the link between AI investment and its outcome. Only four report unconditional evidence. Section 4.2 orders these conditions.

4.2 Synthesis of conditions

Section 4.1 ended on the number that sets this section's task: 63 of the 67 studies identify at least one condition that shapes whether AI investment reaches its outcome. This section orders those conditions into six groups, moving from the investments a firm makes alongside AI to the timing and credibility of the investment itself. The grouping was fixed in a documented clustering of every condition entry in the coding table, and a study appears in several groups when it identifies several conditions. The four studies with unconditional evidence remain the contrast case.

4.2.1 Complementary investments and resources

Eleven of the 63 studies tie the payoff to what a firm invests alongside AI, and this group holds the most direct answers to the question of dose. Among high-tech ventures, revenue reacts only once AI adoption passes a sufficient intensity, and the gains grow where the venture also runs complementary technologies, cloud services and databases among them, and keeps R&D inside the venture (Lee et al., 2022, p. 1). One panel study prices the threshold: the payoff to AI venture funding follows a U-shape whose turning point lies near USD 11.3 million, with short-term declines below that level and long-term gains above it (Banna & Alam, 2025, sec. 4.1).

The complement named most often is the firm's own research activity. In the same panel, AI investment alone stays insignificant while its interaction with R&D turns significantly positive (Banna & Alam, 2025, sec. 4.1). Evidence from UK firms points the same way: AI adoption raises innovation performance where it complements the firm's internal R&D spending, and it partly substitutes collaboration with external knowledge partners (D'Amico et al., 2025, abstract).

Data holds the same position on the archival side. Among Chinese listed firms, data assets carry roughly 45.6 percent of the total value effect of AI adoption (Shi et al., 2025, p. 7). The case evidence describes that dependency from inside the organization: performance arrives only where firms rebuild their processes around the technology, and success rests on a bundle of data, talent, domain knowledge, partnerships and infrastructure (Wamba-Taguimdje et al., 2020, pp. 1893–1894).

The survey studies phrase the same requirement as readiness. Whether Saudi B2B SMEs take up AI practices depends on technology roadmapping, attitude, awareness and infrastructure; professional expertise and technicality stay insignificant (Baabdullah et al., 2021, p. 255). The study behind it treats technological readiness and ICT capability as entry conditions for AI-supported partner management, and both significantly improve its performance (Samadhiya et al., 2023, p. 510). In recruitment, adoption reaches competitive advantage through dynamic capabilities, built from HR competencies together with open innovation; financial backing and IT infrastructure support these capabilities (Sandeep et al., 2025, p. 404).

Two studies extend the complement logic to how the technology is sourced and combined. French firms that source AI externally face more volatile productivity outcomes than in-house developers, and the volatility fades where ICT specialists make up more of the payroll (Fontanelli et al., 2025, pp. 1–2). Pairing AI with robotic process automation adds revenue, mainly through prices, but does not cut costs further, and the revenue gain concentrates where the firm's strategic intent targets growth (Arshad et al., 2026, p. 1). The case studies read the same list from its negative side: data quality, missing skills, high investment needs and unclear economic benefits are the barriers that stand between

AI and supply-chain competitiveness (Cannas et al., 2024, p. 3333). Across the eleven studies of this group, a pattern emerges: none reports that acquiring the technology alone is enough.

4.2.2 Organizational capabilities and human capital as the channel

With 26 of the 63 studies, this is the largest group, and its shared claim is structural: the investment does not act on performance directly, it builds an organizational capability, and the capability delivers the outcome. The claim is strongest where the direct path is empty. Among early-stage ventures, strategic resilience carries the entire link between AI capability and entrepreneurial success, and the authors read the mechanism as organizational, not technological (Wang & Zhang, 2026, p. 1). Italian manufacturers show the same shape: AI reaches manufacturing firm performance through knowledge management processes (Leoni et al., 2022, p. 411). Even value co-creation follows the pattern: its impact on competitive advantage stays insignificant, and the firms' innovation and dynamic capabilities do the transmitting (Y. Sun et al., 2022, p. 474). Adoption lifts organisational performance by way of the quality of decisions and the performance of business processes, with process innovation, automation and organisational learning as partial, complementary mediators (Zebec & Indihar Štemberger, 2024, p. 1). One two-stage survey models AI capabilities as inputs to the firm's adaptive response to market change and finds that this response lifts firm performance and innovation (Sullivan & Fosso Wamba, 2024, p. 1). In accounting firms, AI-infused knowledge pays through green human and structural capital, while the relational channel contributes nothing (Bin-Nashwan & Li, 2025, p. 1). And big-data intelligence raises new product performance by being converted into AI capabilities, with electronic supply-chain collaboration strengthening the indirect path (Tao et al., 2025, abstract).

The dynamic-capabilities studies specify what the channel is made of. Garment exporters hold sustained competitive advantage through sensing the market, seizing it and reconfiguring it, and all three run higher once AI adoption rests on an analytics foundation (Hossain et al., 2022, p. 240). In Malaysian hospitality, sustainable competitive advantage transmits adoption into value, while dynamic capabilities first hold value back during the transition and prove themselves only under turbulence (Alam et al., 2025, p. 1). Among manufacturing SMEs in emerging economies, AI knowledge capabilities work directly and through more radical digitally enabled product innovation, with digital sustainability leadership reshaping both paths (Dinh, 2026, p. 84). In Chinese manufacturing, AI capabilities raise sustainability performance via a tiered mechanism whose prescriptive component fails (Renfei et al., 2026, p. 1).

A cluster of survey studies locates the channel in the marketing function. For food firms, a marketing strategy built on AI raises performance only inside configurations, combined with customer co-creation, marketing capabilities and market orientation (Wu & Monfort, 2023, p. 484). For Chinese cross-border e-commerce SMEs, the pairing of AI adoption with strategic agility strengthens product innovation as well as the upgrading of brands through international marketing capabilities (Guo et al., 2025, p. 1250). Stakeholder engagement built on AI reaches firm performance through marketing agility (Tehrani & Roy, 2025, abstract). CRM systems powered by AI pay through marketing ambidexterity; profitability and competitive advantage follow (Alnofeli et al., 2026, p. 1). For generative AI, technological opportunism converts into competitive advantage through three affordances (Li, Wu, et al., 2026, p. 1).

The human side of the channel is its own line of evidence. The benchmark study of this group states it sharpest: in chess, AI adoption makes the players' traditional strengths obsolete and creates capabilities at the human-machine boundary that have no relation, at times a negative one, to the old ones (Krakowski et al., 2023, p. 1425). Cohesive teams improve how knowledge is shared, and a committed workforce widens the performance gain from adoption (Ullah et al., 2026, p. 1). AI human capital raises firm value partly by compressing wage inequality (Agag et al., 2026, p. 1). AI capabilities reach sustainable SME performance through green innovation and creativity (Alwakid & Dahri, 2025, p. 1). The archival studies find organizational channels as well. Among Chinese listed companies, the key route from AI to productivity is a drop in information asymmetry; more specialized work division and green innovation appear as potential channels (Z. Sun et al., 2024, p. 1). And the retail evidence turns the channel configurational: performance comes from combinations of AI applications rather than single uses, and out of the combinations grow new capabilities, adaptive learning among them (Cao et al., 2025, p. 1). Taken together, the studies of this group agree on the order: the investment builds a capability, the capability earns the return, and several find no significant direct effect at all.

4.2.3 Leadership, governance, and management

Thirteen of the 63 studies condition the payoff on who runs the firm and how it is governed. On the survey side the recurring moderator is leadership. In AI-embedded CRM, leadership support lifts both relationship satisfaction and firm performance (Chatterjee et al., 2022, p. 437). The companion study reports the same moderating role of leadership support for firm performance under AI-CRM (Chatterjee et al., 2021, p. 213). Generative AI adoption lifts firm performance, and ethical leadership moderates that effect (Kumar et al., 2025, p. 1). In hotels and resorts, leader narcissism shapes how the adoption factors translate into competitive positioning (Chakraborty, 2025, abstract). The take-up of AI by key account managers is shaped by their personality traits, and organizational culture moderates the agreeableness path (Mehta et al., 2025, p. 543). On the archival side, innovation gains from AI are amplified where executives hold IT backgrounds (Feng et al., 2025, p. 1).

Managerial capability marks the boundary of the effect in the panel evidence. Firms with a stronger management team pull more value out of the investment (Shi et al., 2025, p. 1). One efficiency study pushes the point furthest: the baseline association between AI investment and efficiency is zero, and gains appear mainly where capable managers, sharper competitive pressure, stable institutional ownership and long-horizon financing come together (Kazakis, 2026, abstract).

The remaining studies move from persons to governance systems. Among Chinese listed companies, part of the AI–TFP link runs through improved control quality (M. Lin et al., 2026, abstract). Investment efficiency gains from AI are more pronounced where knowledge intensity and R&D spending are high and the internal controls sound (N. Lin et al., 2026, p. 1). Responsible AI reaches competitive advantage through supply-chain justice, and complexity erodes the distributive channel while the procedural one holds (Li, Shi, et al., 2026, p. 1). The qualitative pole reports the same dependency: AI-adopting pharmaceutical companies manage tendering faster and decide better, yet run into governance and compliance challenges, and the payoff hinges on aligning adoption with the EU's AI Act and the GDPR (Pesqueira et al., 2024, p. 1). In this group's reading, the same investment returns more under better management and tighter governance.

4.2.4 Firm characteristics and market position

Sixteen of the 63 studies find the payoff sorted by what the firm already is, and size leads the list. In the corpus's benchmark result, growth from AI investment concentrates among the larger firms; industry concentration rises with it (Babina et al., 2024, p. 1). AI-powered pricing shows the same order: the bigger, more productive players are the likelier adopters, and adopters grow faster in sales and employment as well as markups (Adams et al., 2026, p. 1). Hospitals holding more market share are the likelier adopters and improve revenue, occupancy and productivity with AI, with the authors stressing that assessing these gains requires endogeneity controls (Pham et al., 2024, p. 1). Innovation gains cluster in big incumbents and firms in their growth stage (Feng et al., 2025, p. 1). The reverse also holds: when AI services fail, small firms lose more market value than large ones, and accuracy failures cost the most (D. Song et al., 2025, p. 504).

Ownership and financial position sort the effect as well. The productivity gain is largest in state-controlled firms, in internationally active ones, and in innovators (Z. Sun et al., 2024, p. 1). The valuation premium favors state ownership, vanishes in polluting industries unless AI belongs to a green transition that is credible, and appears only where asset structures stay light (Shi et al., 2025, p. 1). Acquisitions after adoption concentrate in firms holding abundant cash, in less developed market environments, and under private ownership (D. Liu et al., 2026, p. 1). Efficiency gains need, next to capable managers, stable institutional ownership and long-horizon financing (Kazakis, 2026, abstract). Investment efficiency responds more where knowledge intensity, R&D spending and internal controls are strong (N. Lin et al., 2026, p. 1). And the market's reward for AI innovation splits along firm traits: return on assets shows a small positive effect at best, stronger for firms heavy on R&D, while firm value rises where sustainability metrics are good, with Democratic administrations amplifying the reward (Singh et al., 2026, abstract).

The downside sorts by the same variables. Investor reactions to AI investment announcements are negative on average, a drop of 1.77 percent on the announcement day, and nonmanufacturing firms, those with a weak IT base, and those with poor credit ratings take the harder hit (Lui et al., 2022, p. 373). Crash risk after AI investment grows where information transparency is low and resources are tight (Bai et al., 2026, p. 1). In Taiwanese semiconductors, innovation from AI lifts sustainability efficiency, while profit efficiency tracks size together with leverage (Chiu et al., 2025, abstract). Industry-4.0 manufacturers convert AI capabilities into sustainability performance far better than traditional ones (Renfei et al., 2026, p. 1). The value association of AI human capital is stronger in travel, the compression of wage inequality stronger in hospitality (Agag et al., 2026, p. 1). Across this group, most sorting runs in one direction: firms that are larger, better resourced, or better rated capture more of the upside and less of the downside.

4.2.5 Environmental conditions

Fifteen of the 63 studies place the condition outside the firm. On the survey side the recurring variable is technological turbulence, and it cuts both ways. It dampens the paths from automated decision making and operational efficiency to relationship satisfaction (Chatterjee et al., 2022, p. 437). In hospitality it moderates selectively, touching only the path from dynamic capabilities to value (Alam et al., 2025, p. 1). On the other side it amplifies: the payoff to AI-based stakeholder engagement grows when technology turbulence is high (Tehrani & Roy, 2025, abstract). Hostile and fast-moving environments condition how much each AI capability feeds the firm's adaptive response (Sullivan & Fosso Wamba, 2024, p. 1). Competition sharpens the payoff as well. Competitive intensity strengthens the creational path from technological opportunism to competitive advantage (Li, Wu, et al., 2026, p. 1). The efficiency gains of the corpus's null-baseline study need, among other conditions, sharper competitive pressure (Kazakis, 2026, abstract).

Regulation can raise or lower the payoff. Where the regulatory climate is clear and its enforcement supportive, the link from AI-enabled innovation to productivity strengthens (Z. Liu et al., 2026, p. 1). Exposure to privacy regulation blunts the gain in investment efficiency (Chen et al., 2026, p. 1). The pharmaceutical tender case declares alignment with the EU's AI Act and the GDPR necessary for the payoff (Pesqueira et al., 2024, p. 1).

Dubai's robotic-surgery advantage required the health authority's approval one case at a time, next to a shift in medical culture (Ali Mohamad et al., 2023, pp. 53, 61). Even partisan politics enters: the market's reward for sustainability-conscious AI innovation grows under Democratic administrations (Singh et al., 2026, abstract).

The macro environment sorts the effect too. The growth payoff of AI investment, 0.04 percent per ten-percent increase in investment, hangs on the labour market: labour productivity works for it, labour costs and the labour share work against it (Tingbani et al., 2025, p. 961). Trade tensions push suppliers into AI integration, and the integration improves efficiency, sales and the firm's market value until extreme exposure to tariffs overwhelms the compensation (Filieri et al., 2026, p. 57). Policy can push the other way: digital-transformation policies along the supply chain add to the productivity gain (Z. Sun et al., 2024, p. 1). Stock market reactions to AI-heavy service patents in manufacturing are negative, softened by idiosyncratic volatility (Patel, 2026, p. 1). What unites this group: the payoff shifts with the environment even when the investment does not.

4.2.6 Dose, credibility, and timing

The last group, fourteen of the 63 studies, conditions the payoff on how much is invested, how credibly it is communicated, and when. Three of its dose findings stand in earlier groups already: the adoption-intensity threshold and the U-shaped funding payoff belong to the complements group (Section 4.2.1), the tariff turning point to the environment group (Section 4.2.5). What this group adds is the shape of the middle: platform embeddedness helps most there, with medium depth doing the most for product innovation and moderate breadth the most for brand upgrading (Guo et al., 2025, p. 1250).

How the investment is deployed carries its own conditions. The case evidence ties performance to firms reworking their processes around AI (Wamba-Taguimdje et al., 2020, p. 1893). Externally bought AI comes with a heightened volatility that in-house development does not show (Fontanelli et al., 2025, pp. 1–2). Pairing AI with robotic process automation adds revenue rather than cost savings, and the gain follows the strategic intent behind the deployment (Arshad et al., 2026, p. 1).

Credibility decides the market's reaction. Announcements of AI implementation earn a positive reaction on the day itself, and the returns differ with the detail of the announcement (Huang & Lee, 2023, p. 9). In corporate filings, actionable disclosures with clear plans bring valuation gains while speculative and irrelevant mentions bring nothing, and silence or vagueness draws penalties (Basnet et al., 2025, p. 1). Unsubstantiated claims, the practice the SEC labels AI washing, first attract investors and then turn into negative reactions and lasting underperformance (X. Song et al., 2026, p. 1).

Time itself conditions the estimates. The productivity effect of AI loses pace over the long run (Z. Sun et al., 2024, p. 1). The negative market impact of AI service failures has grown more intense in recent years (D. Song et al., 2025, p. 504). The negative reaction to AI-heavy service patents shows only in the two most recent decades (Patel, 2026, p. 1). Market value and financial performance rise with implementation, but the edge of first movers and top performers is not uniform across the indicators (Huang & Lin, 2025, p. 1856). Across this group, dose, credibility, and timing each move the payoff.

Four studies of the corpus report unconditional evidence; the other 63 tie the payoff to at least one of the conditions ordered in this section.

Chapter 5

Discussion

This review set out to answer one question: under what conditions do firm-level AI investments translate into competitive advantage and superior firm performance? The preceding chapter delivered the material for the answer; this chapter interprets it.

5.1 The conditional answer

The first half of the answer concerns performance, and it is positive but conditioned. 38 of the 67 studies report a positive average effect, against 20 conditional, five mixed and four negative ones. The positive count alone would overstate the case: 63 of the 67 studies, the positive ones included, identify at least one condition that shapes whether the investment reaches its outcome, and only four report evidence without any. The six groups of Section 4.2 can be read as one sequence: a firm invests alongside the technology, converts the investment into organizational capabilities, steers the conversion through leadership and governance, within what the firm already is and where it operates, at a dose and with a credibility the market accepts. This ordering is the review's own synthesis; no single study tests it end to end.

The conditions rarely arrive alone. In the corpus's null-baseline study, efficiency gains appear where capable managers, competitive pressure, stable institutional ownership and long-horizon financing come together (Kazakis, 2026, abstract). The configurational evidence makes the combination itself the unit of analysis. For food firms, performance follows recipes that join marketing capabilities, customer co-creation, market orientation and an AI marketing strategy (Wu & Monfort, 2023, p. 484). In retail, the payoff grows out of combinations of AI applications rather than any single use (Cao et al., 2025, p. 1). Read this way, the practical question is not which condition matters most. It is which configurations a firm can complete.

The second half of the answer concerns competitive advantage, and it is mostly missing. 15 of the 67 studies measure the construct at all: ten report positive evidence, five conditional. The remaining 52 measure performance and stop there. Barney's test for a sustained advantage, survival after rivals' attempts at duplication have run their course (Barney, 1991, p. 102), is not met by survey scales that capture a perceived edge at one point in time. Across the 15, this review reads only the chess evidence of Krakowski et al. as a test of persistence: the capabilities that emerge at the human-machine boundary stay heterogeneous over time (Krakowski et al., 2023, p. 1425). Performance gains are documented. Sustained competitive advantage is measured rarely, and almost never as persistence.

5.2 Reading the evidence through the four lenses

The general-purpose-technology lens predicted an uneven payoff driven by complements, and the corpus delivers its firm-level picture. Revenue reacts only past an adoption-intensity threshold and grows where complementary technologies and internal R&D stand behind the investment (Lee et al., 2022, p. 1). The funding payoff turns at roughly USD 11.3 million (Banna & Alam, 2025, sec. 4.1). Behind both stands the intangible logic of the productivity J-curve: complements are built, unbooked, before they pay (Brynjolfsson et al., 2021, pp. 333–334). The lag that Brynjolfsson et al. place in the national accounts shows up here as thresholds in firm panels. The lens holds.

The resource-based view holds up against the evidence in its strict reading only. A purchasable technology meets none of Barney's four conditions (Section 2.3), and the corpus behaves accordingly: several studies that model a direct path from investment to outcome find it insignificant (Section 4.2.2). Wang and Zhang state the resulting reading most directly: the mechanism behind AI's performance returns is organizational, not technological (Wang & Zhang, 2026, p. 1). Krakowski et al. place the imitation barrier in the bundle of human and machine capabilities rather than in either part (Krakowski et al., 2023, p. 1446). The firm-characteristics group adds the sorting the RBV expects. Growth from AI concentrates among the larger firms (Babina et al., 2024, p. 1). Weakness sorts the downside the same way: firms with weak IT bases or poor credit ratings take the deeper announcement penalty (Lui et al., 2022, p. 373). What the RBV cannot do is carry the 52 performance-only studies: higher margins show value creation, not inimitability. The construct separation this thesis maintains marks exactly the difference between what the literature has shown and what it has not.

The automation-augmentation distinction enters the evidence as a design condition. Zebec and Indihar Štemberger find no significant direct effect of process automation on process performance once tasks grow complex, and derive the boundary rule: automate the routine, augment the ambiguous (Zebec & Indihar Štemberger, 2024, pp. 16–17). Sourcing adds a corollary: externally bought AI carries a volatility that in-house development does not show (Fontanelli et al., 2025, pp. 1–2). Raisch and Krakowski's argument that the two applications cannot be kept apart in practice reads, after this corpus, like a job description for management. The moderators of Section 4.2.3, from leadership support to managerial capability, describe who draws that line: firms under stronger management pull more value out of the same investment (Shi et al., 2025, p. 1).

Kemp's situated AI fits the overall pattern best, and it is the least tested of the four lenses. The theory holds that a generic technology yields nothing defensible by itself, because rivals can buy and deploy the same thing (Kemp, 2024, p. 620). It gives the firm three activities to work against that: grounding, bounding and recasting (Kemp, 2024, pp. 618–619). The conditionality this corpus documents is what such a theory would predict. The six condition groups can be read as the empirical surface of that claim, with complements and capabilities on the grounding side and governance on the recasting side; this mapping is the review's own proposal, not something the studies test. Direct tests are beginning just outside the corpus. On 570 Spanish manufacturing firms, Vaillant et al. find that firms using AI platforms show more cooperative and capability-building behavior when identifying problems and implementing solutions (Vaillant et al., 2025, abstract). That cooperation leads toward the grounding, bounding and recasting capabilities the theory names, in what the authors call one of the first empirical applications of situated AI (Vaillant et al., 2025, secs. 6.1, 6.2). That study measures no performance outcome and was excluded from this corpus for exactly that reason. As a template for testing Kemp's propositions against outcomes, it is the more valuable.

5.3 Implications for practice

For a firm weighing the investment, the corpus prices the entry ticket. Below the turning point of roughly USD 11.3 million in AI venture funding, performance can decline in the short run (Banna & Alam, 2025, sec. 4.1), and the complements that decide the payoff, from data assets to specialists to reworked processes, cost money the technology budget does not show. The downside is also not symmetric: when AI services fail, small firms lose more market value than large ones (D. Song et al., 2025, p. 504). A firm that cannot fund the complements has, on this evidence, little reason to expect the average positive effect.

Management is a working condition, not a background variable. Across the governance group, the same investment returns more under leadership support, capable management and sound internal controls (Section 4.2.3); none of these arrive with the software. The capability channel of Section 4.2.2 adds the sequencing: the investment builds a capability first and earns its return second, which means the payoff arrives only after the organizational work is done.

Communication about the investment carries payoff conditions of its own. Actionable disclosure with clear plans earns valuation gains while speculative mentions earn nothing (Basnet et al., 2025, p. 1). Rhetoric that outruns implementation is punished: AI washing draws investors in at first, and what follows are negative market reactions and lasting underperformance (X. Song et al., 2026, p. 1).

5.4 Limitations

Four limitations bound these conclusions. The first is composition: 33 of the 67 studies measure AI through survey constructs and self-reports, and many of their outcomes are perceived rather than accounted. The second is identification, and most of the corpus is associational. The hospital evidence makes the point from inside the corpus: assessing AI's payoff requires endogeneity controls (Pham et al., 2024, p. 1). The third is coverage: samples from the United States and China make up 47.8 percent of the corpus, and two thirds of the studies (65.7 percent) appeared in 2025 and 2026, so long-run outcomes are largely unobservable, and the one study that decomposes the effect over time finds it slowing already (Z. Sun et al., 2024, p. 1). The fourth is the review protocol itself: a single database, journal articles in English, a journal-quality threshold of AJG 2, and a query scoped to investment rather than any AI use, all documented in Chapter 3. A corpus assembled this way can only contain what journals publish, and null results are the finding least likely to reach publication; the four-study unconditional contrast group should be read with that in mind.

5.5 Future research

The largest gap follows from the construct separation: competitive advantage needs to be measured as persistence, not as a perceived edge. 15 of 67 studies measure the construct, and almost all of them capture it cross-sectionally; designs that follow adopters and their imitators over years, in the way the chess panel follows players, would show whether any AI-based edge survives duplication. Kemp's theory offers the propositions and now has a first operationalization to build on; connecting Vaillant et al.'s situating capabilities to performance and persistence outcomes is the obvious next step. The dose findings deserve replication outside venture funding, since a threshold that moves with firm size or industry would change the practical advice. The make-or-buy margin, so far documented once for volatility, deserves the same attention for performance levels. Coverage should widen beyond the two economies that dominate the samples. And the corpus's newest layer, generative AI, is still measured at adoption; its performance evidence has yet to accumulate.

Chapter 6

Conclusion

This thesis asked under what conditions firm-level AI investments translate into competitive advantage and superior firm performance, and answered it with a systematic review of 67 empirical studies published between 2020 and 2026, drawn from a documented Scopus search of 432 records, screened in three recorded stages, and coded by two independent coders against a frozen scheme. The coding table in the appendix is the review's core deliverable; every number in the text derives from it.

The answer favors conditions over the investment itself. Across the corpus, 38 of the 67 studies report positive average effects, and the evidence is conditional almost throughout: 63 studies identify at least one condition that decides whether the investment reaches its outcome. The review ordered these conditions into six groups, from the complementary investments a firm makes alongside the technology to the credibility with which the investment is communicated. For competitive advantage, the evidence is thin. 15 studies measure the construct, almost always as a perceived edge at one point in time, and only a single study in the corpus follows an advantage against imitation over time.

The contribution is the ordered map itself. Where individual studies report single moderators and mediators, the review shows that the conditions recur in patterned groups across methods and settings, that the configurational evidence finds them working in combination rather than alone, and that the two outcome constructs the field often blends carry different amounts of evidence. The map marks where the next studies are needed, with persistence the least covered and most consequential claim. Firms can read the same map as a checklist of what has to be in place before the investment can be expected to pay.

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Appendix

Coding table

The full coding table for all 67 included studies, split into study characteristics (Table A.1) and findings (Table A.2).

Table A.1: Included studies: characteristics and design (n = 67). Source: author's compilation.

ID	Study	Journal (AJG)	Country	Sample	Method	AI investment measure
S01	Wamba-Taguimdje et al. (2020)	Business Process Management Journal (2)	multi-country	~500 vendor-published AI project mini-cases (IBM, AWS, Cloudera, Nvidia etc.), archival qualitative analysis	case study	case observation (AI-based transformation projects from vendor websites)
S02	Baabdullah et al. (2021)	Industrial Marketing Management (3)	Saudi Arabia	392 B2B SMEs, cross-section, CB-SEM	survey-SEM	survey construct (acceptance of AI practices)
S03	Chatterjee et al. (2021)	Industrial Marketing Management (3)	India	349 managers from 16 BSE-listed firms (11 manufacturing, 5 service), response rate 51.6%, PLS-SEM	survey-SEM	survey construct (AI-CRM implementation)
S04	Chatterjee et al. (2022)	Journal of Business Research (3)	India	312 usable manager responses from 14 BSE-listed firms (10 large/4 SME; telecom, IT, automobile, retail, pharma), Jan-Feb 2020, RR 47.3%, PLS-SEM	survey-SEM	survey construct (adoption of AI-embedded CRM system)
S05	Hossain et al. (2022)	Industrial Marketing Management (3)	Bangladesh	257 usable manager responses (1,953 approached, 278 qualified), RMG export manufacturers, research-firm panel/Qualtrics, random sampling; time-lagged robustness subsample n=93	survey-SEM	survey construct (AI adoption on top of marketing analytics platform)
S06	Lee et al. (2022)	Technovation (3)	South Korea	160 high-tech ventures (from 1,248 ministry list, 300 responses), survey + audited financials	panel econometrics	survey (AI-adoption intensity) linked to financial records
S07	Leoni et al. (2022)	International Journal of Operations and Production Management (4)	Italy	120 senior executives of manufacturing firms, PLS-SEM	survey-SEM	survey construct (AI adoption/use)
S08	Lui et al. (2022)	Annals of Operations Research (3)	multi-market	119 AI investment announcements by 62 listed firms, Factiva search Jan 2015 - Feb 2019	event study	announcements (AI investment announcements)

Table A.1 (continued)

ID	Study	Journal (AJG)	Country	Sample	Method	AI investment measure
S09	Mishra et al. (2022)	Journal of the Academy of Marketing Science (4*)	USA	19,000 firm-year observations, US-listed firms (COMPUSTAT + EDGAR 10-Ks), simultaneous equations; PSM robustness (1,801 treatment / 2,251 control firms)	panel econometrics	10-K text (share of AI keywords = AI focus)
S10	Sun et al. (2022)	Systems Research and Behavioral Science (2)	China	234 AI-related manufacturers (GM/CEO respondents), CB-SEM	survey-SEM	survey construct (value co-creation in AI innovation ecosystem)
S11	Ali et al. (2023)	Journal of Organizational Change Management (2)	UAE	single case (CMC Dubai): 9 semi-structured interviews with robotic-surgery team + archival data, triangulated	case study	case observation (adoption of AI/robotic surgery)
S12	Czarnitzki et al. (2023)	Journal of Economic Behavior and Organization (3)	Germany	5,851 firms (409 AI users, ~7%), German CIS 2018 (ZEW); cross-section OLS/2SLS-IV + panel subset with IV-FE, entropy balancing	panel econometrics	survey (CIS AI adoption dummy + intensity of AI use in business processes)
S13	Huang & Lee (2023)	Data Base for Advances in Information Systems (2)	USA	109 AI implementation announcements 2014-2019, S&P 500 components (Compustat) + Google Search event collection	event study	announcements (AI implementation)
S14	Krakowski et al. (2023)	Strategic Management Journal (4*)	multi-country	same chess players across conventional, centaur and engine tournaments (controlled competitive setting)	panel econometrics	AI (engine) adoption in competitive interaction
S15	Samadhiya et al. (2023)	Industrial Marketing Management (3)	India	in-depth interviews with senior B2B managers + PLS-SEM on 142 responses	mixed	survey construct (AI-based partner relationship management implementation)
S16	Wu & Monfort (2023)	Psychology and Marketing (3)	Taiwan	278 food firms (CEO/marketing managers), Taiwan, survey Mar-Aug 2020, RR 33%; SEM + supplementary fsQCA	survey-SEM	survey construct (implementation of AI marketing strategy)
S17	Babina et al. (2024)	Journal of Financial Economics (4*)	USA	1,993 US Compustat firms; resume-based AI workforce measure (Cognism), robust to job-postings measure; IV = university AI graduate supply	panel econometrics	resume-based (share of AI-skilled workers as firm AI investment)

Table A.1 (continued)

ID	Study	Journal (AJG)	Country	Sample	Method	AI investment measure
S18	Cannas et al. (2024)	International Journal of Production Research (3)	Italy	multiple case study: 6 companies, 17 AI implementation cases, semi-structured interviews	case study	case observation (AI applications in SCOR processes)
S19	Pesqueira et al. (2024)	Computers and Industrial Engineering (2)	Europe	comparative 4 pharma companies (2 AI adopters vs 2 non-adopters) + 2-round Delphi with 53 participants	mixed	adoption status (AI in tender management)
S20	Pham et al. (2024)	Journal of Business Research (3)	USA	941 AI hospital-year obs + 941 matched non-AI hospital-year obs (246 matched hospitals), 40 US states, 2000-2020	panel econometrics	adoption dummy (hospital AI adoption)
S21	Sullivan & Fosso (2024)	Journal of Business Research (3)	USA	two-wave survey, 107 US executives completing both waves (of 225 wave-1; 47.5% effective RR), panel-recruited IT+business executives	survey-SEM	survey constructs (AI-enabled automation, AI-enabled analytics, AI-enabled relational capabilities)
S22	Sun et al. (2024)	Technology in Society (2)	China	3,235 Chinese listed firms, 2007-2021; fixed effects + system GMM + mediation models	panel econometrics	annual-report text (frequency of AI-related terms); controls from CSMAR/RESSET
S23	Zebec & Indihar (2024)	Business Process Management Journal (2)	EU	448 organisations, EU, serial multiple mediation SEM	survey-SEM	survey construct (AI adoption)
S24	Alam et al. (2025)	International Journal of Hospitality Management (3)	Malaysia	336 respondents, stratified purposive sampling, hospitality industry	survey-SEM	survey construct (AI adoption + technology readiness)
S25	Alwakid & Dahri (2025)	Technology in Society (2)	Saudi Arabia	250 SMEs (managers/owners), manufacturing + services	survey-SEM	survey construct (AI capabilities: infrastructure, business integration, proactive stance)
S26	Banna & Alam (2025)	International Journal of Entrepreneurial Behaviour and Research (3)	EU	unbalanced panel of 1,479 firms (~12,900 obs), 26 EU countries, 2012-2023; CGM multi-way clustering + 2SLS-IV	panel econometrics	AI-related venture-capital funding received (log), OECD AI Policy Observatory data (turning point USD 11.3M); robustness: AI job intensity, AI patents, industry-level IV
S27	Basnet et al. (2025)	International Review of Financial Analysis (3)	USA	US firms, 10-K filings 2005-2018 (pre-hype window); AI mentions classified actionable/speculative/irrelevant; causal design	panel econometrics	10-K text (actionable vs speculative vs irrelevant AI disclosures)

Table A.1 (continued)

ID	Study	Journal (AJG)	Country	Sample	Method	AI investment measure
S28	Bin-Nashwan & Li (2025)	Technology in Society (2)	China	433 valid responses (461 screened, 5000+ invitations), Chinese accounting professionals, online survey Dec 2023-Jan 2024, PLS-SEM	survey-SEM	survey construct (AI-infused knowledge systems)
S29	Cao et al. (2025)	European Management Review (3)	USA	Study 1: instrument from 6,519 AI documents of 37 retailers; Study 2: fsQCA on survey of 140 U.S.-based retail managers	fsQCA	validated multi-level instrument (AI integration at task/function/firm level)
S30	Chakraborty (2025)	Current Issues in Tourism (2)	India	two-wave online survey, India: 1,487 invited, wave 1 n=437, wave 2 n=408 (hotel/resort managers, multi-channel recruitment)	survey-SEM	survey construct (GenAI adoption)
S31	Chiu et al. (2025)	Business Ethics the Environment and Responsibility (2)	Taiwan	33 publicly listed semiconductor firms; three-stage DEA (profit, sustainability, market) + Tobit regressions	DEA	patents (AI-driven innovation proxied by patent activity)
S32	D'Amico et al. (2025)	Journal of Technology Transfer (3)	UK	14,143 UK firms, 2004-2020	panel econometrics	archival: Beauhurst-identified AI technology/product use, matched to Business Structure Database + UK Innovation Survey (2004-2020, biennial)
S33	Feng et al. (2025)	Quarterly Review of Economics and Finance (2)	China	4,004 Chinese A-share firms, 2011-2023; procurement-based AI adoption index + patents + supply-chain data	panel econometrics	procurement index (AI adoption from procurement records)
S34	Fontanelli et al. (2025)	Journal of Economic Behavior and Organization (3)	France	French ICT survey 2019, firm-level; balancing/matching robustness	panel econometrics	survey (predictive AI use; buyers vs in-house developers)
S35	Guo et al. (2025)	International Marketing Review (3)	China	322 SMEs using cross-border platforms	survey-SEM	survey construct (AI adoption coupled with strategic agility)
S36	Huang & Lin (2025)	Managerial and Decision Economics (2)	USA	S&P 500 firms: 50 AI adopters (23 first movers, 27 followers) identified via 97 manually screened AI-implementation news items + non-adopter controls; Compustat 2010-2017, baseline 2017	panel econometrics	news announcements (manually verified AI implementation)

Table A.1 (continued)

ID	Study	Journal (AJG)	Country	Sample	Method	AI investment measure
S37	Kumar et al. (2025)	Journal of Business Research (3)	multi-country	Study 1: in-depth interviews; Study 2: SEM on 277 B2B managers across USA, UK, Canada, India, Australia, Malaysia, Japan	survey-SEM	survey construct (GenAI adoption)
S38	Mehta et al. (2025)	Journal of Business and Industrial Marketing (2)	India	Phase 1: 26 key-account-manager interviews (automobile); Phase 2: survey n=496, SEM	mixed	survey construct (adoption of AI technologies by key account managers)
S39	N. & Roy (2025)	European Journal of Marketing (3)	Australia	335 companies across industries, PLS-SEM	survey-SEM	survey construct (AI-based stakeholder engagement)
S40	Sandeep et al. (2025)	Journal of Intellectual Capital (2)	India	290 HR professionals in talent acquisition, purposive sampling, South India, PLS-SEM	survey-SEM	survey construct (AI adoption in recruitment)
S41	Shi et al. (2025)	International Review of Financial Analysis (3)	China	Chinese listed companies 2010-2022 (~22,900 firm-year obs); Heckman correction, endogeneity tests	panel econometrics	annual-report text (Word2vec-based AI dictionary) + secondary measure
S42	Song et al. (2025)	Industrial Management and Data Systems (2)	USA	120 AI service failure events 2010-2024 at NYSE/NASDAQ-listed companies (from 451 initial events)	event study	announcements (AI service FAILURE events)
S43	Tao et al. (2025)	Journal of Business and Industrial Marketing (2)	China	291 B2B manufacturing firms, multi-wave multi-source, SEM	survey-SEM	survey construct (big data analytical intelligence assimilation + AI capabilities)
S44	Tingbani et al. (2025)	International Journal of Finance and Economics (3)	USA	1,950 unique US firms, 1996-2016, Compustat; system GMM	panel econometrics	resume/job-based AI workforce measure (Fedyk & Hodson 2023 approach)
S45	Adams et al. (2026)	Journal of Monetary Economics (4)	USA	US firms matched to Lightcast job postings 2010Q1-2024Q1 (40,000+ job boards); AI-pricing jobs identified	panel econometrics	job postings (AI-skill + pricing keyword = AI pricing adoption)
S46	Agag et al. (2026)	Tourism Management (4)	UK	1,206 UK listed tourism/travel/hospitality firms, 7,539 firm-years 2018-2024; two-step system GMM	panel econometrics	AI human capital (AI-skilled workforce measure)
S47	Alnofeli et al. (2026)	International Journal of Information Management (2)	Australia	3-stage: scoping review + expert interviews -> instrument; survey of 205 banking employees, SEM	survey-SEM	survey construct (AI-powered CRM capability, higher-order)

Table A.1 (continued)

ID	Study	Journal (AJG)	Country	Sample	Method	AI investment measure
S48	Arshad et al. (2026)	International Journal of Information Management (2)	multi-country	two survey datasets of large firms worldwide (RPA + AI implementation)	survey-based regression (two cross-sectional surveys)	survey (joint implementation of RPA and AI)
S49	Bai et al. (2026)	Pacific Basin Finance Journal (2)	China	Chinese A-share listed firms; quasi-natural experiment (New-generation AI Pilot Zone Policy); pre-registered	panel econometrics	policy shock (AI pilot zone exposure) as exogenous AI investment driver
S50	Chen et al. (2026)	Journal of Empirical Finance (3)	USA	US firms; AI mentions in earnings calls, validated vs 10-K; IV = textual proximity to university AI research; 2SLS	panel econometrics	earnings-call text (share of AI terminology in management remarks)
S51	Dinh (2026)	Industrial Marketing Management (3)	Vietnam	sequential mixed-methods, 261 manufacturing SMEs	mixed	survey construct (AI-enabled knowledge capabilities)
S52	Filieri et al. (2026)	Journal of Product Innovation Management (4)	China	Chinese listed manufacturers + WTO tariff records + customs data 2015-2022; IV analysis	panel econometrics	archival AI integration measure (tariff-induced)
S53	Giordino et al. (2026)	Technological Forecasting and Social Change (3)	Europe	balanced panel, 432 European listed companies 2015-2023 (LSEG data), banks/insurers excluded	panel econometrics	organizational AI focus (archival, LSEG-based)
S54	Kazakis (2026)	Scottish Journal of Political Economy (2)	USA	US Compustat firms; efficiency via DEA (normalized ratio scores); labor-based AI investment measure	panel econometrics	labor-based AI investment measure
S55	Li et al. (2026)	International Journal of Production Economics (3)	China	218 firms in supply chains	survey-SEM	survey construct (responsible AI adoption)
S56	Li et al. (2026)	Journal of Business Research (3)	China	223 firms across industries, China, survey	survey-SEM	survey construct (GenAI affordances: organizational memory, collaborative, creational)
S57	Lin et al. (2026)	Emerging Markets Finance and Trade (2)	China	Chinese A-share listed companies, 2007-2022	panel econometrics	annual-report text-based AI adoption measure (authors explicitly contrast with patent/robot proxies)
S58	Lin et al. (2026)	International Review of Financial Analysis (3)	China	Chinese listed firms	panel econometrics	tailored AI-keyword dictionary, novel text analysis of company reports
S59	Liu et al. (2026)	International Review of Financial Analysis (3)	China	Chinese publicly listed firms, 2007-2024	panel econometrics	textual analysis of annual reports (CNINFO, jieba segmentation), ln(1+AI keyword frequency), following Mishra et al. 2022

Table A.1 (continued)

ID	Study	Journal (AJG)	Country	Sample	Method	AI investment measure
S60	Liu et al. (2026)	Technology in Society (2)	Asia-Pacific	Study 1: multilevel survey of 420 firms (HLM); Study 2: vignette experiment with 360 managers	mixed	survey construct (AI-enabled innovation); experimentally manipulated regulatory conditions
S61	Mukherjee et al. (2026)	International Journal of Quality and Reliability Management (2)	India	312 valid responses from luxury hotels (lodging + food services)	survey-SEM	survey construct (AI adoption)
S62	Patel (2026)	Technological Forecasting and Social Change (3)	USA	1,306 AI-scored service patents (of 758k patents) from 3,899 manufacturing firms, 1980-2020; matched-pair robustness	event study	patents (AI score of service technology patents)
S63	Renfei et al. (2026)	Technovation (3)	China	292 valid responses (295 distributed, two-stage collection; pretest 100/85), 120 Chinese manufacturing enterprises, PLS-SEM	survey-SEM	survey construct (AI capabilities: perceptive, predictive, prescriptive)
S64	Singh et al. (2026)	Business Strategy and the Environment (3)	USA	S&P 500 companies, 2000-2024; system-GMM + PSM robustness	panel econometrics	AI patents granted per year, identified via PATENTSCOPE AI Index methodology (1 SD = 115.66 patents -> Tobin Q +0.058, ~3% firm value)
S65	Song et al. (2026)	Finance Research Letters (2)	USA	US firms 2018-2023; BERT-classified AI disclosures vs AI patent portfolios	panel econometrics	text-vs-patents gap (unsubstantiated AI claims = AI washing)
S66	Ullah et al. (2026)	International Journal of Information Management (2)	China	419 manufacturing firms; SEM + LSTM validation	survey-SEM	survey construct (AI adoption)
S67	Wang & Zhang (2026)	International Journal of Information Management (2)	China + Europe	two-wave survey of 357 early-stage ventures (China + Europe); PLS-SEM + IPMA + fsQCA + interviews	mixed	survey construct (AI innovation capability)

Table A.2: Included studies: outcomes, effect directions, and conditions (n = 67). Source: author's compilation.

ID	Study	Outcome	Outcome measure(s)	Direction	Conditions	Key finding
S01	Wamba-Taguimdje et al. (2020)	Perf.	Perf: organizational performance (financial, marketing, administrative) + process-level performance, qualitative	conditional	process reconfiguration required (mechanism: performance gains only when AI features are used to reconfigure processes); AI as configuration of complements (data, talent, domain knowledge, partnerships, infrastructure)	AI transformation projects improve organizational and process performance, but only when firms use AI to reconfigure their processes rather than overlay it on existing ones.
S02	Baabdullah et al. (2021)	Perf.	Perf: perceived SME performance (survey scale)	positive	technology roadmapping (enabler, sig.); attitude (sig.); infrastructure readiness (sig.); awareness (sig.); professional expertise and technicality not significant	Acceptance of AI practices improves perceived SME performance; adoption is driven by roadmapping, attitude, infrastructure and awareness - not by professional expertise.
S03	Chatterjee et al. (2021)	Both	Perf: perceived organizational performance (survey scale); CA: competitive advantage scale (COA1-3, Rogers-based; incl. market-share gain), discriminant-valid vs OP; predicted by performance	positive	mediators: B2B engagement, employee experience, information processing capability (all sig.); moderator: leadership support (sig., MGA); firm size/age/industry only controls (not moderators)	AI-CRM implementation improves perceived organizational performance and competitive advantage in B2B relationship management.
S04	Chatterjee et al. (2022)	Perf.	Perf: firm performance scale (market+financial criteria, ROI/ROA, sales growth, market share, profitability) + B2B relationship satisfaction	positive	technology turbulence weakens the paths from automated decision-making and operational efficiency to B2B relationship satisfaction (negative moderator); leadership support strengthens the path from satisfaction to firm performance (positive moderator); the main effect of AI-CRM on firm performance is unconditional	AI-embedded CRM improves relationship satisfaction and firm performance; gains shrink under technology turbulence and grow with leadership support.
S05	Hossain et al. (2022)	CA	CA: sustained competitive advantage = relative performance vs competitors over last 3 years (market share growth, sales growth, profitability, ROI)	conditional	mediators: market sensing, seizing and reconfiguring carry the effect of marketing analytics capability on sustained competitive advantage in full; moderator: AI adoption strengthens all three capability paths	Marketing analytics capability drives sensing/seizing/reconfiguring toward sustained competitive advantage; AI adoption amplifies this only when built on an analytics foundation.
S06	Lee et al. (2022)	Perf.	Perf: revenue growth	conditional	AI-intensity threshold (low-level adoption: no effect); complementary technology investments (cloud, database) amplify; internal/exclusive R&D strategy amplifies	AI pays off only above an adoption-intensity threshold, and more so with complementary technology investments and venture-specific internal R&D.

Table A.2 (continued)

ID	Study	Outcome	Outcome measure(s)	Direction	Conditions	Key finding
S07	Leoni et al. (2022)	Perf.	Perf: perceived manufacturing firm performance (survey scale)	conditional	full mediation through knowledge management processes: the direct path from AI to manufacturing firm performance is not significant while both legs of the indirect path are; AI has no direct effect on supply chain resilience either; larger firms reach higher AI maturity	AI alone does not improve manufacturing firm performance - the effect exists exclusively through knowledge management processes (full mediation).
S08	Lui et al. (2022)	Perf.	Perf: firm market value (CAR; -1.77% on event day)	negative	nonmanufacturing firms (more negative); weak IT capability (more negative); low credit rating (more negative)	investors react negatively to AI investment announcements on average; the penalty concentrates in nonmanufacturing firms with weak IT capability and low credit ratings.
S09	Mishra et al. (2022)	Perf.	Perf: gross and net operating efficiency, net profitability, adspend, employment	mixed		AI focus reduces gross operating efficiency but increases net operating efficiency and profitability - automation raises costs more slowly than revenues.
S10	Sun et al. (2022)	CA	CA: perceived competitive advantage scale (own construct, Table 3) + innovation intelligibility; no performance construct in model	conditional	full mediation: neither value co-creation directly nor the economic and relationship value it creates significantly affects competitive advantage; dynamic capabilities and innovation capabilities carry the entire effect	Value co-creation in AI ecosystems has no direct effect on competitive advantage - effects run entirely through dynamic and innovation capabilities.
S11	Ali et al. (2023)	Both	Perf: four qualitative outcome categories: clinical (errors/infections down, patient experience), financial (quantified: ~500K/month at 30 robotic cases), organizational (resource allocation), technological; CA: competitive position with explicit competitor comparison: patient choice over competitors, bargaining power with third-party payers, differentiation, upper hand over competition (P1)	positive	qualified robotic team incl. revenue-cycle team (capability prerequisite); cultural shift in medical community; regulatory approval (DHA, case-by-case); high acquisition costs not covered by insurance (entry barrier)	AI-enabled robotic surgery improves clinical, financial and technological outcomes and strengthens the competitive position of the healthcare organization.

Table A.2 (continued)

ID	Study	Outcome	Outcome measure(s)	Direction	Conditions	Key finding
S12	Czarnitzki et al. (2023)	Perf.	Perf: firm productivity (labor- and value-added-based)	positive		AI adoption is robustly associated with higher firm productivity in representative German data; the effect grows with usage intensity.
S13	Huang & Lee (2023)	Perf.	Perf: firm market value (positive abnormal returns on event day)	positive	announcement detail (detailed > vague); IT vs non-IT firms (sig. difference); frequency of negative words in announcement (negative correlation)	Markets reward AI implementation announcements on average - more so when announcements are detailed and come from IT firms.
S14	Krakowski et al. (2023)	Both	Perf: competitive performance (game outcomes/ratings); CA: sources of persistent performance heterogeneity (= sustained advantage) across settings	conditional	substitution: traditional human capabilities become obsolete; complementation: new human-machine capabilities emerge that are unrelated or negatively related to traditional ones	AI adoption shifts the sources of competitive advantage - a new decision-making resource emerges at the human-AI intersection, unrelated to traditional capability.
S15	Samadhiya et al. (2023)	Both	Perf: perceived firm performance (survey scale); CA: perceived sustainable competitiveness (survey scale)	positive	ICT capability (prerequisite); technological readiness (prerequisite); firm fit	AI-based partner relationship management improves firm performance and sustainable competitiveness, provided ICT capability and technological readiness are in place.
S16	Wu & Monfort (2023)	Perf.	Perf: perceived firm performance (survey scale)	positive	fsQCA configurations: marketing capabilities + customer value co-creation + market orientation + AI strategy jointly form necessary and sufficient recipes for high performance	AI marketing strategy raises firm performance, but only as part of configurations with marketing capabilities, co-creation and market orientation - a configurational result.
S17	Babina et al. (2024)	Perf.	Perf: growth in sales, employment, and market valuation	positive	channel: product innovation rather than process efficiency; gains concentrate among firms that were already large, which raises industry concentration (superstar dynamics)	AI-investing firms grow in sales, employment and value via product innovation; benefits concentrate among large firms, increasing concentration.
S18	Cannas et al. (2024)	Perf.	Perf: qualitative competitiveness outcomes: costs, lead times, service level, quality, safety, sustainability	positive	barriers as boundary conditions: data quality, AI skills, high investment needs, unclear economic benefits, lack of AI cost-analysis experience	AI improves supply-chain competitiveness across SCOR processes, but realization depends on overcoming data-quality, skill and investment-evaluation barriers.

Table A.2 (continued)

ID	Study	Outcome	Outcome measure(s)	Direction	Conditions	Key finding
S19	Pesqueira et al. (2024)	Both	Perf: operational efficiency in tender management (processing times, decision quality, document accuracy) - expert-rated/qualitative; CA: expert-rated competitiveness dimensions per company (Delphi, Likert+IQR); bid competitiveness in tendering context	conditional	governance and ethical frameworks (contingency, per conclusions); regulatory compliance (EU AI Act, GDPR); organizational readiness; senior management involvement (identified critical factor); alignment with business goals	AI adopters manage pharmaceutical tendering faster and with better decisions than non-adopters, but only sustainably within robust governance and compliance structures.
S20	Pham et al. (2024)	Perf.	Perf: outpatient revenue, inpatient revenue, productivity, occupancy	positive	market share (larger-share hospitals adopt AI and benefit); endogeneity control essential - naive estimates misleading	Hospitals with larger market share adopt AI and gain in revenue, productivity and occupancy; without endogeneity controls the performance effects are misestimated.
S21	Sullivan & Fosso (2024)	Perf.	Perf: firm performance + process innovation + product innovation (survey scales)	conditional	adaptive response to market changes is a full mediator, with no direct AI-to-performance paths modelled; environmental hostility weakens the automation path (negative moderator) and turns the relational path insignificant; environmental dynamism moderates as well; only ten of eighteen conditional indirect effects are significant	AI capabilities pay off through the firms adaptive response to market changes; environment hostility/dynamism condition how much each AI capability contributes.
S22	Sun et al. (2024)	Perf.	Perf: firm total factor productivity	positive	stronger in state-controlled, internationally oriented, innovative firms; mechanism: reduced information asymmetry (key channel), specialized division of labor, green innovation; supply-chain digitalization policy amplifies; effect slows over the long run	AI raises firm productivity in China - most for state-controlled, international and innovative firms - but the productivity boost decays over time.
S23	Zebec & Indihar (2024)	Perf.	Perf: organizational performance (survey scale)	positive	serial mediation: decision-making quality and business-process performance carry the effect; process automation, organizational learning, process innovation as complementary partial mediators	AI creates business value indirectly: through better decisions and business-process performance, complemented by automation, learning and process innovation.

Table A.2 (continued)

ID	Study	Outcome	Outcome measure(s)	Direction	Conditions	Key finding
S24	Alam et al. (2025)	Both	Perf: value creation (survey scale); CA: sustainable competitive advantage (survey scale, modeled as mediator)	positive	sustainable competitive advantage mediates between AI adoption and value creation; dynamic capabilities first impede value creation through transition costs and become crucial later under turbulence; technological turbulence moderates only the path from capabilities to value creation	AI adoption creates value in hospitality mainly through sustainable competitive advantage as transmission channel; dynamic capabilities help only once transition costs are absorbed.
S25	Alwakid & Dahri (2025)	Perf.	Perf: sustainable SME performance (survey scale)	positive	SME creativity and green innovations (mediators); green entrepreneurial orientation as parallel driver (risk-taking, innovativeness, proactiveness)	AI capabilities raise sustainable SME performance through creativity and green innovation - AI infrastructure plus proactive strategy is the entry condition.
S26	Banna & Alam (2025)	Perf.	Perf: firm revenue growth	conditional	U-shaped effect with a quantified turning point at roughly USD 11.3M of AI venture funding: below it AI drags revenue, above it AI pays; coupling AI with an R&D innovation strategy amplifies the effect, standalone AI is insufficient	AI investment follows a U-shaped payoff: short-term revenue drag, long-term gains - and only firms coupling AI with R&D strategy capture the upside.
S27	Basnet et al. (2025)	Perf.	Perf: firm value (market valuation); R&D spending, patents, lagged productivity as channels	conditional	what counts is the substance of the disclosure: actionable disclosures bring valuation gains, innovation and lagged productivity, speculative or irrelevant ones bring nothing, and silent or vague peers are penalised	Markets reward only substantive (actionable) AI disclosures - early adopters with real implementation plans gain value via innovation; vague AI talk earns nothing.
S28	Bin-Nashwan & Li (2025)	Perf.	Perf: accounting-firm performance: efficiency, accuracy, compliance, reporting innovation, timeliness (survey)	conditional	no direct path from AI-infused knowledge to performance is modelled; green human capital and green structural capital carry the effect, the green relational capital channel is empty; sustainability culture moderates the path from green intellectual capital to performance	AI-infused knowledge improves accounting-firm performance through green human and structural capital, amplified by sustainability culture; relational capital plays no role.
S29	Cao et al. (2025)	Perf.	Perf: efficiency and innovation outcomes (configurational)	conditional	configurational: no single AI function suffices - synergistic combinations (esp. customer service + cybersecurity with other functions); emergent capabilities: adaptive learning, predictive analytics, uncertainty mitigation	Retail performance comes from configurations of AI-enabled functions, not isolated AI uses - a directly configurational (fsQCA) result.

Table A.2 (continued)

ID	Study	Outcome	Outcome measure(s)	Direction	Conditions	Key finding
S30	Chakraborty (2025)	Both	Perf: firm/organisational performance (survey scales); CA: competitive advantage / competitive positioning (survey scale)	positive	technological readiness (TOE driver); leader narcissism moderates adoption-factor relationships and GAI effects on competitive positioning	GenAI adoption improves hotels competitive advantage and performance, with leadership personality (narcissism) shaping how adoption translates into outcomes.
S31	Chiu et al. (2025)	Perf.	Perf: stage efficiencies: profitability, ESG/sustainability, market valuation	mixed	stage-specific: AI innovation positive on sustainability + market stage efficiency, insignificant on profit stage (size/leverage drive profit); firm scale raises profit efficiency but lowers sustainability/market efficiency (asymmetry)	AI-driven innovation lifts sustainability and market-valuation efficiency in semiconductors but not short-term profit efficiency - the payoff is stage-specific.
S32	D'Amico et al. (2025)	Perf.	Perf: innovation performance (knowledge spillover of innovation)	conditional	complementarity with own R&D investment (AI pays only if it complements internal R&D); AI substitutes knowledge collaboration with certain external partners	AI adoption boosts innovation only when paired with the firms own R&D investment - and partly replaces external knowledge collaboration.
S33	Feng et al. (2025)	Perf.	Perf: innovation output and innovation efficiency	positive	stronger for large incumbents and growth-stage firms; executives with IT backgrounds amplify; highly innovative firms diversify supply chains to reduce resource risk	AI adoption raises innovation output and efficiency, most where firm maturity, scale and IT-savvy leadership provide absorptive conditions.
S34	Fontanelli et al. (2025)	Perf.	Perf: volatility of productivity growth (not the level)	conditional	AI buyers: higher volatility; in-house developers: none; complementary human capital (share of ICT engineers/technicians) mitigates the buyer effect	Off-the-shelf AI raises the riskiness of productivity outcomes; developing in-house or holding complementary ICT human capital neutralizes it.
S35	Guo et al. (2025)	CA	CA: international advantage: product innovation + brand upgrading (GVC position, survey scales)	positive	international marketing capabilities (mediator); platform embeddedness depth (inverted U on product innovation) and breadth (inverted U on brand upgrading) - medium embeddedness optimal	AI adoption coupled with strategic agility upgrades SMEs global value-chain position through marketing capabilities - but only at moderate platform embeddedness (inverted-U).
S36	Huang & Lin (2025)	Perf.	Perf: financial ratios (Compustat), productivity, market value (Tobin q)	mixed	indicator-specific: financial performance + market value positive, productivity insignificant; first movers show partly negative/insignificant effects (no uniform first-mover advantage)	AI implementation lifts financial performance and market value, but first movers and top performers do not benefit uniformly across all indicators.
S37	Kumar et al. (2025)	Perf.	Perf: perceived firm performance (survey scale)	positive	ethical leadership moderates the path from GenAI adoption to performance; adoption reasons for and against it (uniqueness, information completeness, convenience, deceptiveness)	GenAI adoption boosts perceived firm performance, more strongly under ethical leadership.

Table A.2 (continued)

ID	Study	Outcome	Outcome measure(s)	Direction	Conditions	Key finding
S38	Mehta et al. (2025)	Both	Perf: firm performance (survey scale); CA: competitive advantage (survey scale, modeled as mediator)	positive	manager personality traits (Big Five) drive adoption; competitive advantage mediates between adoption and performance; organizational culture moderates the effect of agreeableness on adoption	Individual personality traits shape AI adoption in key account management, which converts into firm performance through competitive advantage - culture conditions the path.
S39	N. & Roy (2025)	Perf.	Perf: firm performance (survey scale)	positive	marketing agility (mediator); technological turbulence amplifies the positive effects (high-turbulence environments gain more)	AI-based stakeholder engagement improves performance through marketing agility - and the payoff is largest in technologically turbulent environments.
S40	Sandeep et al. (2025)	CA	CA: perceived competitive advantage from AI-enabled recruitment (survey scale)	positive	HR competencies and open innovation build dynamic capabilities, which mediate; financial support and IT infrastructure act as enabling resources	AI adoption in recruitment yields competitive advantage only where dynamic capabilities, built from HR competencies and open innovation with financial/IT support, are in place.
S41	Shi et al. (2025)	Perf.	Perf: enterprise value (Tobins Q)	positive	data assets mediate 45.6% of the effect while a significant direct effect remains; managerial capability moderates; boundary conditions: no effect in heavily polluting industries unless the firm has a credible green transition, and none in asset-intensive firms; state-owned enterprises gain about twice as much as private ones	AI adoption raises enterprise value through data assets, and only under capable management and agile (non-asset-heavy) structures - the study explicitly maps when AI becomes a source of advantage.
S42	Song et al. (2025)	Perf.	Perf: firm value (negative abnormal returns)	negative	small firms hit harder; recent years more negative; failure-type order effect: accuracy > safety > privacy > fairness	AI service failures destroy market value - most for small firms and accuracy failures; the downside risk of AI implementation is real and priced.
S43	Tao et al. (2025)	Perf.	Perf: new product performance (survey scale)	positive	AI capabilities (mediator: BDAI assimilation alone insufficient); electronic supply-chain collaboration (moderator, strengthens the indirect effect)	Big-data/AI assimilation lifts new product performance only when converted into AI capabilities, and more so with electronic supply-chain collaboration.
S44	Tingbani et al. (2025)	Perf.	Perf: firm growth (sales growth)	conditional	labour market conditions: labour productivity amplifies the growth effect, while labour cost and labour share weaken it (a ten percent rise in AI investment adds 0.04% growth on average)	AI investment raises firm growth modestly, and the payoff hinges on labour-market conditions - productivity-rich, labour-cost-light firms capture it.

Table A.2 (continued)

ID	Study	Outcome	Outcome measure(s)	Direction	Conditions	Key finding
S45	Adams et al. (2026)	Perf.	Perf: growth in sales, employment, assets, markups; stock-return response to monetary policy	positive	adoption concentrated in larger, more productive firms; adopters more responsive to monetary policy surprises	Firms adopting AI-powered pricing grow faster in sales, employment and markups - AI pricing is a capability large productive firms exploit first.
S46	Agag et al. (2026)	Perf.	Perf: firm value; wage inequality as mediating channel	positive	wage inequality partially mediates: AI human capital compresses inequality and thereby raises firm value; the effect is stronger in dynamic and complex industries; by sector, the value effect is strongest in travel and the inequality compression in hospitality	AI human capital raises firm value partly by compressing wage inequality - a workforce-centered pathway to AI-enabled advantage, strongest in dynamic sectors.
S47	Alnofeli et al. (2026)	Both	Perf: profitability (survey scale); CA: competitive advantage (survey scale)	positive	marketing ambidexterity (mediator: exploration/exploitation balance carries the effect)	AI-CRM capability lifts profitability and competitive advantage through marketing ambidexterity - capability without ambidexterity closes the value-realisation gap only partially.
S48	Arshad et al. (2026)	Perf.	Perf: revenue growth (via price increases); cost reduction tested	mixed	component split: RPA and AI jointly add revenue growth through price increases but no additional cost reduction (the joint term is insignificant, while each technology on its own does cut costs); a strategic revenue-growth intent amplifies the revenue outcomes	Combining RPA and AI pays through revenue enhancement, not cost efficiency - and only for firms whose strategic intent targets growth.
S49	Bai et al. (2026)	Perf.	Perf: stock price crash risk (up) + firm value (up)	mixed	crash risk rises via reduced information transparency + managerial optimism; worse for low-transparency and resource-constrained firms; firm value gains may be short-term market optimism	Policy-driven AI investment inflates firm value but simultaneously raises crash risk - short-term valuation gains trade off against long-term stability.
S50	Chen et al. (2026)	Perf.	Perf: corporate investment efficiency; Tobins q (long horizon)	positive	mechanisms: better sales forecasts, reporting quality, process/product innovation; data-privacy regulation exposure attenuates the effect	AI adoption improves how firms allocate capital - unless regulatory friction (data-privacy exposure) blunts it; value shows up in Tobins q only over longer horizons.
S51	Dinh (2026)	Perf.	Perf: new product performance (survey scale)	positive	dual pathway: direct + via digital product innovation radicalness; digital sustainability leadership as both mediator and moderator (reconfigures how AI creates value)	AI knowledge capabilities lift new product performance in emerging-market SMEs, with digital sustainability leadership as the critical boundary condition.

Table A.2 (continued)

ID	Study	Outcome	Outcome measure(s)	Direction	Conditions	Key finding
S52	Filieri et al. (2026)	Perf.	Perf: supply chain efficiency, sales performance, market value	conditional	inverted U: AI performance benefits diminish at extreme tariff exposure ("optimal investment threshold"); customer engagement needs strengthen the effect (boundary condition); tariff exposure induces AI integration	Trade tensions push firms into AI integration that improves efficiency, sales and value - up to a threshold where adversity overwhelms the compensation.
S53	Giordino et al. (2026)	Perf.	Perf: ROA + Tobins Q (+ ESG pillar scores)	positive		Organizational AI focus goes hand in hand with better financial performance and better E/S sustainability scores in European listed firms.
S54	Kazakis (2026)	Perf.	Perf: firm efficiency	conditional	no unconditional effect; gains only with: capable managers, stronger competitive pressure, stable institutional ownership, long-term debt access	AI investment alone does nothing for efficiency - gains materialize exclusively under managerial, competitive, ownership and financing conditions ("Conditional Gains").
S55	Li et al. (2026)	CA	CA: perceived competitive advantage (survey scale)	conditional	full mediation: the direct path from responsible AI to competitive advantage is not significant, the effect runs entirely through distributive and procedural justice; supply chain complexity weakens the distributive path but not the procedural one	Responsible AI builds competitive advantage through supply-chain justice - in complex networks, only procedural fairness remains a robust channel.
S56	Li et al. (2026)	CA	CA: perceived sustained competitive advantage (survey scale)	positive	three GenAI affordances mediate between technological opportunism and competitive advantage; competitive intensity amplifies only the creational-affordance path	Technological opportunism converts into competitive advantage through GenAI affordances - under intense competition, only the creational affordance keeps paying.
S57	Lin et al. (2026)	Perf.	Perf: total factor productivity	positive	internal control quality partially mediates: AI improves governance and risk management, which in turn raises total factor productivity	AI raises firm productivity partly by improving internal control quality - governance is a transmission channel, not just a moderator.
S58	Lin et al. (2026)	Perf.	Perf: investment efficiency + firm value	positive	stronger in high knowledge-intensity, high R&D-intensity firms and firms with sound internal controls; mechanisms: innovation capability + financial health	AI enhances investment efficiency and firm value most where knowledge intensity, R&D and internal controls provide absorptive capacity.
S59	Liu et al. (2026)	Perf.	Perf: M&A likelihood (strategic outcome) + long-term firm value	positive	stronger with abundant cash holdings, less-developed market environments, private ownership	AI adoption makes firms more acquisitive and raises long-term value - internal resources and institutional context set the boundary.

Table A.2 (continued)

ID	Study	Outcome	Outcome measure(s)	Direction	Conditions	Key finding
S60	Liu et al. (2026)	Perf.	Perf: productivity gains + firm growth	positive	the regulatory climate moderates how strongly innovation translates into productivity, with clarity and supportive enforcement jointly maximising it (moderated mediation); an experiment in study 2 provides causal evidence for the regulatory conditions	AI-enabled innovation converts into productivity and growth only under clear, supportive regulation - institutional environment is the amplifier.
S61	Mukherjee et al. (2026)	Perf.	Perf: service performance (survey scale)	positive		AI adoption lifts hotel service performance where ease of use, competitive pressure, infrastructure and policy support jointly create momentum.
S62	Patel (2026)	Perf.	Perf: stock market reaction to patent grants (negative abnormal returns)	negative	idiosyncratic volatility mitigates the negative reaction; tangibility does not; effects only in 2010s/2020s (evolving market perception)	Markets react negatively to AI-heavy service patents in manufacturing - a caution against assuming short-term market value from AI servitization.
S63	Renfei et al. (2026)	Perf.	Perf: corporate sustainability performance incl. explicit economic dimension (EP01-EP04: cost reduction, waste revenue, spin-offs)	positive	prescriptive capabilities fail as a component of the AI capability bundle; firm type is a boundary condition: the effect is strong for Industry-4.0 manufacturers and significantly weaker, though not absent, for traditional firms	AI capabilities raise sustainability (incl. economic) performance mainly in Industry-4.0-ready manufacturers - prescriptive AI fails, and traditional firms barely benefit.
S64	Singh et al. (2026)	Perf.	Perf: ROA + firm value	conditional	small positive ROA effect, stronger for R&D-intensive firms; firm-value gains only for firms with good sustainability metrics; stronger under Democratic administrations (political environment as condition)	AI innovation pays modestly and unevenly: R&D intensity, sustainability credentials and even the political climate condition whether markets reward it.
S65	Song et al. (2026)	Perf.	Perf: market reactions + long-term operational performance	negative	time path: short-lived positive investor reaction, then negative BHAR + sustained underperformance; substantiation gap (claims vs patents) is the treatment	AI talk without AI substance backfires: markets initially reward the narrative, then punish rhetoric that outpaces implementation.
S66	Ullah et al. (2026)	Perf.	Perf: organizational performance (survey scale)	positive	team dynamics, corporate innovation capabilities, high-commitment workforce (mediators/amplifiers of the AI-performance link)	AI adoption converts into performance where cohesive teams, innovation capability and a committed workforce absorb it - human/organizational enablers unlock the value.

Table A.2 (continued)

ID	Study	Outcome	Outcome measure(s)	Direction	Conditions	Key finding
S67	Wang & Zhang (2026)	Perf.	Perf: entrepreneurial success (venture performance, survey scale)	conditional	full mediation: strategic resilience fully accounts for the relationship between AI capability and venture success; fsQCA identifies equifinal configurations that lead to high performance	AI capability makes ventures successful only via organizational resilience - the mechanism is organizational, not technological; multiple configurations lead there.

List of Aids

The following AI tools and aids were used during the preparation of this thesis.

Global (all sections)

Section	**Tool**	**Description**
ALL	Grammarly Pro	For mistake and style checking throughout the thesis. Mistakes were corrected immediately; style suggestions were reviewed manually and applied when suitable.
ALL	Claude Code (Anthropic)	Used throughout for LaTeX text correction and conversion, literature management (renaming PDFs, managing BibTeX), file structure, and drafting/editing section content. All AI-generated text was reviewed and approved before inclusion.

Section	Tool	Description
ALL (writing phase)	library-rag — local retrieval tool over the thesis corpus (built with Claude Code, 27 July 2026)	Local search system over the 67 full texts of the frozen corpus, used from the writing phase onward to locate citable passages. Runs entirely on the author's machine, no cloud services: Docling for PDF parsing, Qwen/Qwen3-Embedding-4B (fp8) for semantic search, BAAI/bge-reranker-v2-m3 for reranking, hybrid semantic and BM25 retrieval fused by reciprocal rank fusion. The tool retrieves and locates passages; it does not generate thesis text. Every verbatim quote it returns passes a two-stage programmatic check before it may enter a footnote: the quote must appear verbatim in the retrieved source passage, and its words must appear in the same order on the cited page of the PDF itself. The second stage also catches a quote attributed to the wrong passage or page. Its threshold was calibrated against 40 genuine and 60 fabricated quotes from this corpus (genuine: median 1.00; fabricated: maximum 0.18; the threshold of 0.60 admits none of the fabricated ones). Quotes the check cannot confirm are flagged for manual inspection of the page rather than passed through. Printed page numbers are calibrated per document from the printed page labels in the PDF and cross-checked against the Scopus page range; where no printed page number exists, the tool states this instead of supplying one. Corpus identification, screening, and coding were completed and frozen (17–18 July 2026) before this tool existed and are unaffected by it.

Section	Tool	Description
ALL (writing phase, from 13 Aug 2026)	Antigravity CLI (Google)	Successor backend for the blind citation checks: Google discontinued the Gemini CLI sign-in for individual accounts (it still worked on 28 July 2026 and was refused by 13 August 2026), so corpus/gemini_verify_citations.py now calls the Antigravity CLI (model: Gemini 3.1 Pro) instead of Gemini 2.5 Pro. The blind design is unchanged — the model still receives nothing but the sentence and the footnote passage, and the tool now additionally runs in an empty working directory so it cannot read any project file. Each verdict record notes the model that produced it; all verdicts from before the switch remain Gemini 2.5 Pro and are labelled as such. The migrated script was verified with a deliberately overclaiming test citation, which the new model flagged correctly.
Proposal	Claude Code (Anthropic)	Readability rewrite pass over the proposal prose; minimal simplifications merged into proposal_final.tex after review; citations, quotes, and footnotes left unchanged.
Proposal	Quillbot Paraphraser	Comparison paraphrase of citation-free proposal sentences (proposal_quillbot.tex) to evaluate phrasing alternatives; citations, quotes, and footnotes left unchanged.
Proposal	Claude Code (Anthropic)	Source check of the proposal against the collected article PDFs: every cited claim compared with the source text, page numbers and verification footnotes added to all citations, and five wordings corrected where they had drifted from the source. All corrections reviewed before merging.
Proposal	Claude Code (Anthropic)	Shortening pass after I found the draft too dense: text condensed to 3 pages, one main statement per source, detail sentences removed. Citations and quotes kept verbatim; result reviewed by me.

Section	**Tool**	**Description**
Method	Claude Code (Anthropic)	Corpus retrieval (17 July 2026): wrote and ran the script corpus/pull_corpus.py, which executes the documented Scopus query via the Scopus API (WU license) and saves the frozen export (432 records, raw JSON + CSV). Verified that all three benchmark studies are retrieved and that record counts match Scopus.
Method	Claude Code (Anthropic)	Title/abstract pre-screening (17 July 2026): Claude read the full abstracts of all 174 studies passing the AJG $\geq$ 2 filter and proposed include/exclude decisions with reason codes against the criteria catalogue approved beforehand; borderline cases flagged for full-text checking. All proposals recorded in corpus/screening_2026-07-17.csv and reviewed by the author before becoming final.
Method	Claude Code (Anthropic)	Full-text screening assistance (17–18 July 2026): Claude extracted and summarized method/measurement sections from the PDFs of all flagged borderline studies and proposed verdicts against the criteria catalogue; 12 studies excluded content-based, 1 confirmed as include; two inaccessible reports excluded as not retrieved (author's decision). Claude also matched, renamed, and de-duplicated the downloaded PDFs. All verdicts recorded with reasons and reviewed by the author.

Section	**Tool**	**Description**
Results (coding)	Claude Code (Anthropic) + Gemini CLI (Google)	Dual independent AI coding of all 67 included studies against the frozen coding scheme (18 July 2026): Claude drafted all rows from the full texts (coder 1); Gemini 2.5 Pro coded the same studies blind—receiving only the coding scheme and PDF text, never Claude's drafts (coder 2). A documented self-consistency pass harmonized coder 1's effect-direction codes before comparison. Intercoder agreement: method 90%, outcome construct 88%, effect direction 73%.
Results (coding)	Claude Code (Anthropic)	Adjudication support (18 July 2026): for all 29 coder disagreements and 10 random-sample agreement rows, Claude performed full-text reads and prepared evidence briefs quoting both coders' positions with page references; 15 factual gaps resolved from the full texts. The author adjudicated all disagreements against these briefs (18 individually, the remainder in a final batch review); decisions recorded per case, decision rules documented as written case law. No joint coder errors found in the sample checks; all 67 rows finalized after the author's batch approval.

Section	Tool	Description
Theory (Section 2.1)	Claude Code (Anthropic) + library-rag	Drafting of Section 2.1 on the construct distinction (28 July 2026): citable passages located with the local retrieval tool; all four citations checked two-stage (verbatim in the source passage, word sequence on the cited PDF page) and their printed page numbers confirmed (pp. 215, 247, 550). Extended on 28 July 2026 by Barney's (1991, p. 102) definitions of competitive advantage and sustained competitive advantage, taken from the verified text sidecar and read off the page image; the corpus retrieval tool does not cover sources outside the corpus. The counts come from the fact sheet derived from the coding table, not from the tool. BibTeX entries generated from the frozen Scopus export with corpus/make_bib.py. Human-voice pass applied. Text reviewed and edited by the author before inclusion.
Results (descriptive mapping)	Claude Code (Anthropic)	Descriptive figures (18 July 2026): wrote and ran the script corpus/make_figures.py, which derives five corpus-mapping charts (publication year, method, sample geography, outcome construct × effect direction, AJG rating) directly from the finalized coding table and writes them to figures/ as PDF. Category groupings are documented in the script; all counts verified to sum to $n = 67$. Colors validated as colorblind-safe. Figures reviewed by the author before inclusion.

Section	Tool	Description
Theory (Sections 2.2–2.5)	Claude Code (Anthropic) + library-rag	Drafting of the four theory sections on general-purpose technology, the resource-based view, automation/augmentation and situated AI (28 July 2026). Citable passages from the corpus located with the local retrieval tool; all 11 corpus quotes checked two-stage (verbatim in the source passage, word sequence on the cited PDF page). Two quotes flagged by the check were inspected manually on the page; one revealed a page error (Krakowski et al., 1444 → 1445), which was corrected. The three theory anchors outside the corpus (Kemp 2024, Raisch & Krakowski 2021, Brynjolfsson et al. 2021) are not in the retrieval index; their passages were read directly from the PDFs with pdfplumber and the printed page numbers verified against the running heads. Barney (1991) is cited from the verified text sidecar to the image scan (literature/barney1991firm.txt, OCR base checked against the scan page by page); both quoted passages were additionally read off the page images before the footnotes were set, and the original's printed typos are reproduced verbatim. BibTeX entries generated from the frozen Scopus export with corpus/make_bib.py. Human-voice pass applied, including the paraphrase check against every footnote passage. Text reviewed and edited by the author before inclusion.

Section	Tool	Description
Theory (Sections 2.1–2.5)	Claude Code (Anthropic) + Gemini CLI (Google)	Blind citation check of the finished theory chapter (28 July 2026), continuing the dual-AI principle from the coding phase. Claude wrote the script corpus/gemini_verify_citations.py, which parses every citation site in a section and splits the check by what can be decided mechanically: whether a quote appears verbatim on the cited page is settled deterministically by the local retrieval tool, the source-echo rule (three or more consecutive words shared between the prose and the cited passage) is computed by the script, and only the judgement of paraphrase fidelity goes to Gemini 2.5 Pro — blind, receiving nothing but the sentence and the footnote passage. Of 25 citation sites in the chapter, the script flagged 7 for shared word sequences; the verdicts are recorded per site and collected in a report. The author decides every flag; the script proposes nothing and changes no text.

Section	Tool	Description
Method / bibliography	Claude Code (Anthropic)	Citation rule for the seven studies without printed page numbers (28 July 2026): each of the seven checked against its publisher type and the rule approved by the author implemented — article number in the bib eid field plus the article-internal page for the three Elsevier articles, section references for the four with publisher-relative or not-yet-assigned pagination. Two defects in bib.bib were found and fixed in the process: an unescaped ampersand in one title (would have aborted the LaTeX run at the first citation of it) and acronyms being lowercased by APA sentence casing, which had printed "ai", "sme" and "b2b" in the reference list. corpus/make_bib.py now escapes ampersands and brace-protects acronyms, with self-tests; the 27 existing entries were corrected in one pass and the reference list checked in the compiled PDF. One Scopus metadata error corrected against the source PDF: the author of the European Journal of Marketing article was recorded as "N., N. A." and is Tehrani, A. N.
Results (descriptive mapping)	Claude Code (Anthropic)	Drafting of Section 4.1 (18 July 2026): Claude drafted the descriptive-mapping prose directly from the finalized coding table (aggregate counts only, no source citations) and applied the agreed human-voice pass (banned-pattern audit, voice matching). All numbers cross-checked against the coding table; text reviewed and edited by the author before inclusion.

Section	**Tool**	**Description**
Appendix (coding table)	Claude Code (Anthropic)	Reader-facing cleanup of the appendix coding table (29 July 2026): a rendering layer in corpus/make_appendix_table.py strips coder shorthand from the printed table — path coefficients, significance stars, hypothesis labels, model-fit values, and coder notes removed mechanically from 26 cells; 23 cells whose internal path abbreviations could not be resolved by rule were reworded into plain sentences, stored separately in corpus/appendix_overrides.tsv. The coded values in corpus/coding_table.csv are untouched — the layer changes wording only, never a category, effect direction, or the identity of a condition. Two self-checks run at every regeneration: no shorthand token may survive into the rendered table, and every number printed must appear verbatim in the coded source cell. All 23 rewordings documented side-by-side (coded vs. appendix wording) for the author's review.

Section	Tool	Description
Method	Claude Code (Anthropic)	Revision of the Method chapter following the author's review of the interim draft (13 August 2026), worked through question by question with the author: the AJG filter step marked as requiring no reading (ISSN matching), the full-text stage reworded to "assessed for eligibility" with the check explained, AI assistance in screening disclosed in the chapter text, the search documented as run through the Scopus API, the coding-field list split into two explained halves, all five effect-direction values and the two coding rules written out with examples, and two example citations added for the performance/competitive-advantage distinction (Chatterjee et al. 2021, p. 209; Babina et al. 2024, p. 1) — both footnote passages read from the PDFs and the printed page number verified against the running head. Every change decided by the author.
Method, Theory, Results	Claude Code (Anthropic)	AI-pattern audit (13 August 2026) over all drafted chapters against the Wikipedia "Signs of AI writing" catalogue: four wording corrections (an emphasis adverb and a three-item example list in Method, a built antithesis in Theory, a stock idiom in Results) plus one cross-chapter consistency fix (Results now states that effect directions cover four of the five direction codes defined in Method). Citations, quotes, footnotes, and all counts unchanged; factual enumerations deliberately left as they are. Corrections listed for the author's review.

Section	Tool	Description
Method	Claude Code (Anthropic) + Antigravity CLI (Google)	Blind citation check of the finished Method chapter (13 August 2026), first run on the migrated backend: all 3 citation sites (Rojon et al. 2021; Chatterjee et al. 2021; Babina et al. 2024) checked mechanically for source echo (0 hits) and by Gemini 3.1 Pro for paraphrase fidelity, blind. The first run flagged 2 sites; both traced to sentence-initial citation placement, which pairs the footnote with the wrong sentence — the two example sentences were restructured to the claim-then-cite convention. The re-run flagged the Chatterjee site once more, correctly: the footnote covered only the definition half of the sentence, so the H4 passage (same page) was added and the unsupported "three-item" detail dropped. Final result: 3 of 3 sites pass both checks.
Method	Claude Code (Anthropic) + Antigravity CLI (Google)	Blind section review of the finished Method chapter (13 August 2026): Gemini 3.1 Pro received only the chapter and the fact sheet and flagged 5 findings (2 consistency, 3 style). The author decided all five: 3 fixes applied (the count of 29 adjudicated disagreements clarified as studies with at least one disagreement across the five compared fields; "unconditional average effect" reworded to "sign of the average effect" to remove a collision with the fact sheet's conditions counts; one "not only … but also" construction removed), 2 flags rejected as substantive prose. Decisions recorded per flag.

Section	Tool	Description
Results (synthesis)	Claude Code (Anthropic) + Antigravity CLI (Google)	Conditions clustering for the structure of Section 4.2 (13 August 2026): Claude extracted all condition entries from the finalized coding table (63 studies) and proposed six thematic clusters with per-study assignments, flagging 8 judgment cases. Gemini 3.1 Pro classified the same 8 cases blind (cluster definitions and verbatim condition entries only, no proposals). Agreement on 4 cases; the author decided all 8: one Gemini addition adopted, three rejected with reasons, the single genuine dissent resolved for consistency with the theory chapter. Assignments frozen; cluster counts are script-derived and validated against the coding table.
Results (synthesis, Section 4.2.1)	Claude Code (Anthropic) + Antigravity CLI (Google)	Drafting of the Section 4.2 opening and Subsection 4.2.1 on complementary investments (13 August 2026), covering the 11 studies of the frozen cluster. The local retrieval tool was unavailable in this session, so all citable passages were read directly from the source PDFs with pdfplumber; 25 quoted passages across the 11 studies were verified programmatically before entering footnotes (word sequence of every quote confirmed on the cited PDF page; printed page numbers taken from the page headers, section references for the two studies without usable pagination). Corpus counts in the prose come from the fact sheet only. The blind citation check (Gemini 3.1 Pro) ran three times during drafting: the first pass flagged 10 of 12 citation sites, leading to three precision corrections in the prose and footnote extensions with the studies' sample statements; the final pass leaves one flag, the unavoidable word-sequence match on the quantified threshold "USD 11.3 million", for the author's decision. Human-voice pass applied during drafting. Text to be reviewed by the author.

Section	Tool	Description
Results (synthesis, Section 4.2.2)	Claude Code (Anthropic) + Antigravity CLI (Google)	Drafting of Subsection 4.2.2 on organizational capabilities as the channel (13 August 2026), the largest cluster: 22 of its 26 studies are cited individually. All passages read directly from the source PDFs with pdfplumber; 42 quoted passages verified programmatically before entering footnotes (word sequence on the cited PDF page); printed page numbers from the review index's per-document offsets, abstract references for the two studies with unusable pagination. Sixteen BibTeX entries generated from the frozen Scopus export. The blind citation check (Gemini 3.1 Pro) ran three times: the first pass flagged 17 of 34 sites, leading to five precision corrections in the prose and sample-statement extensions in seven footnotes; the final pass leaves five flags, all word-sequence echoes on construct names, sample descriptors, or the quantified threshold, each with Gemini judging the paraphrase fully supported — for the author's batch decision. Text to be reviewed by the author.

Section	Tool	Description
Results (synthesis, Section 4.2.3)	Claude Code (Anthropic) + Antigravity CLI (Google)	Drafting of Subsection 4.2.3 on leadership, governance, and management (13 August 2026): 12 of the cluster's 13 studies cited individually. Seventeen quoted passages verified programmatically on the cited PDF pages before entering footnotes; one wording assumption was falsified by the check and corrected to the page's actual wording. Printed pages from page headers and the review index; abstract references for the three studies without usable pagination. Eight BibTeX entries generated from the frozen Scopus export. The blind citation check (Gemini 3.1 Pro) ran three times: seven initial flags led to four precision corrections in the prose and three footnote extensions with sample statements. Final state across the whole chapter: 9 of 46 citation sites flagged, all word-sequence echoes on construct names, sample descriptors, directive names, or the quantified threshold, each judged fully supported by Gemini; five already decided by the author, four awaiting the author's batch decision.

Section	Tool	Description
Results (synthesis, Section 4.2.4)	Claude Code (Anthropic) + Antigravity CLI (Google)	Drafting of Subsection 4.2.4 on firm characteristics and market position (13 August 2026): all 16 studies of the cluster cited individually, including the two benchmark studies Babina et al. and Lui et al. Nineteen quoted passages verified programmatically on the cited PDF pages; seven BibTeX entries generated from the frozen Scopus export, three proper nouns brace-protected by hand. The blind citation check (Gemini 3.1 Pro) ran three times: thirteen initial flags led to ten precision corrections in the prose (claims cut back to what the abstracts state) and one footnote extension. Final state across the whole chapter: 15 of 62 citation sites flagged, every one a word-sequence echo on construct, sample, or metric names with Gemini judging the paraphrase fully supported; nine decided by the author (recorded durably), six awaiting the author's batch decision.
Results (synthesis, Section 4.2.5)	Claude Code (Anthropic) + Antigravity CLI (Google)	Drafting of Subsection 4.2.5 on environmental conditions (13 August 2026): all 15 studies of the cluster cited individually, five re-cited with passages already verified in earlier subsections. Eighteen quoted passages verified programmatically on the cited PDF pages; six BibTeX entries generated from the frozen Scopus export. The blind citation check (Gemini 3.1 Pro) ran four times: ten initial flags led to seven precision corrections in the prose (two multi-study sentences split, two narrative frames removed, three wordings tightened) and six footnote extensions. Final state across the whole chapter: 22 of 77 citation sites flagged, every one a word-sequence echo on construct, sample, directive, or metric names with Gemini judging the paraphrase fully supported; fifteen decided by the author, seven awaiting the author's batch decision.

Section	Tool	Description
Results (synthesis, Section 4.2.6)	Claude Code (Anthropic) + Antigravity CLI (Google)	Drafting of Subsection 4.2.6 on dose, credibility, and timing (13/14 August 2026), completing Section 4.2: all 14 studies of the cluster cited individually, six re-cited with previously verified passages. Twelve quoted passages verified programmatically; one extraction artifact resolved by reading the printed page image directly. Four BibTeX entries generated from the frozen Scopus export. The blind citation check (Gemini 3.1 Pro) ran three times: six initial flags led to five precision corrections and two footnote extensions. Final state for the finished chapter: 27 of 91 citation sites flagged, all word-sequence echoes on construct, sample, or term names with Gemini judging every paraphrase fully supported; fifteen decided by the author, twelve awaiting the author's batch decision.
Results (Chapter 4)	Claude Code (Anthropic) + Antigravity CLI (Google)	Blind section review of the finished Results chapter (14 August 2026): Gemini 3.1 Pro received only the chapter prose and the fact sheet and flagged 4 findings — one genuine consistency error (two subsections restated the same dose findings nearly verbatim; fixed by cross-referencing instead of repeating) and three style patterns (a section-closing formula repeated four times, reduced to one instance; two stock idioms replaced). The author approved all fixes. Final state: 88 citation sites, 26 flagged, all word-sequence echoes with Gemini judging every paraphrase fully supported, all covered by the author's recorded decisions. Chapter 4 thereby passed all three cross-check gates.

Section	Tool	Description
Theory (Sections 2.1–2.5)	Claude Code (Anthropic) + Antigravity CLI (Google)	Citation re-check of the theory chapter on the migrated backend (14 August 2026): alongside the 7 known word-sequence flags, the migrated script surfaced the paraphrase-fidelity verdicts, including one genuine misreading (Barney 1991, p. 106 — the prose had inverted the passage's point that the VRIN characteristics alone do not make an attribute a resource). Nine precision corrections applied to the prose and four footnotes extended with sample or mechanism passages read from the PDFs (printed pages verified against running heads). Final state: 8 of 25 citation sites flagged — 6 word-sequence echoes on construct or term names with the paraphrase judged fully supported, one definition-not-finding note, one simplification of Barney's fourth condition — all decided by the author in one batch session (14 August 2026, recorded durably). The theory chapter thereby passed all three cross-check gates.
Theory (Sections 2.1–2.5)	Claude Code (Anthropic) + Antigravity CLI (Google)	Blind section review of the theory chapter (14 August 2026, Gate 3 of the cross-check workflow): Gemini 3.1 Pro received only the chapter and the fact sheet and flagged 2 findings, no number contradictions. One consistency flag accepted and fixed (a transition promised an illustration of the productivity dip but led into a study finding a clear gain; reworded to announce the contrast the paragraph actually draws), one style flag rejected as void (the flagged sentence is a verbatim source quote inside a footnote). Both resolutions confirmed by the author in the same batch session.

Section	Tool	Description
Appendix (coding table)	Claude Code (Anthropic)	Full-text verification of all appendix abbreviation resolutions (14 August 2026): every acronym resolution in the 23 reworded cells checked against its source PDF. Nineteen confirmed exactly; four precision corrections applied (Leoni's construct completed to "manufacturing firm performance", Sullivan's mediator to "adaptive response to market changes", Bin-Nashwan's moderator path to "green intellectual capital", and the Sun 2022 cell extended to name the two null value paths the rewording had silently dropped). Table regenerated with its self-checks passing.
Discussion	Claude Code (Anthropic) + Antigravity CLI (Google)	Drafting of the full Discussion chapter (14 August 2026): research question answered separately for the two constructs, evidence interpreted through the four theory lenses, implications, limitations, future research. 25 citation sites, of which 22 re-cite previously verified passages and 3 cite Vaillant et al. (2025), the first empirical application of situated AI theory, read directly from the PDF (early view, section-based citations). The blind citation check ran three times: 18 initial flags led to six sentence splits, four footnote extensions, and six precision corrections (causal wording weakened to the sources' correlational findings, theory claims separated from corpus findings, own-synthesis framing separated from cited claims). Final state: 10 of 25 sites flagged, all word-sequence echoes on construct, sample, or threshold names judged fully supported, all decided by the author in one batch session (14 August 2026, durable log). All corpus counts from the fact sheet.

Section	Tool	Description
Discussion	Claude Code (Anthropic) + Antigravity CLI (Google)	Blind section review of the Discussion chapter (14 August 2026, Gate 3; footnotes stripped before submission): Gemini 3.1 Pro flagged 7 findings — no number contradictions, no consistency errors, seven style patterns. Six fixes applied; one flag rejected because the parallelism is the source's own summary rule. All resolutions confirmed by the author (14 August 2026). The Discussion chapter thereby passed all three cross-check gates.
Introduction	Claude Code (Anthropic) + Antigravity CLI (Google)	Drafting of the Introduction (14 August 2026) along the four mandated moves: Kemp's theoretical puzzle as the anchor (two new passages read column-wise from the PDF), prior research summarized through four already-verified corpus passages, the research question stated explicitly, the outlook covering protocol and chapter structure. Citation check ran twice (two genuine fixes: a footnote extension for the sample descriptor, association wording restored at Babina; remaining flags are fully supported term echoes). Blind section review flagged 3 findings, all fixed: search-window wording disambiguated, two antithesis constructions rewritten plainly. All resolutions and flags decided by the author (14 August 2026). The Introduction thereby passed all three cross-check gates.

Section	Tool	Description
Conclusion + Abstract	Claude Code (Anthropic) + Antigravity CLI (Google)	Drafting of the Conclusion chapter and the Abstract (14 August 2026), both citation-free summaries with all counts from the fact sheet. The blind section review ran over both jointly and flagged 4 findings, all fixed — including one genuine number-attribution error (corpus-wide counts phrased as construct-specific ones), a combinations claim cut back to the configurational evidence, and two antithesis constructions rewritten plainly. All four resolutions reviewed individually and accepted by the author (14 August 2026).
Method (interim draft)	Claude Code (Anthropic)	First supervisor draft (29 July 2026): Claude assembled the standalone compile of the interim draft — TOC skeleton, the full Method chapter, and the coding-table excerpt (the 25 studies in the highest-ranked journals plus three AJG 2 examples). Citation-verification footnotes are suppressed in the compiled draft (working files keep them); the drafted theory chapter and the List of Aids were left out of that round — the former to keep the review load at one chapter, the latter pending the supervisor's guidance on how AI use should be documented.

Section	Tool	Description
Method	Claude Code (Anthropic)	Drafting of the full Method chapter (29 July 2026): protocol prose assembled from the frozen method documentation (query wording, run date, AJG $\geq$ 2 rule, screening chain 432 $\rightarrow$ 174 $\rightarrow$ 81 $\rightarrow$ 67, E-code catalogue, dual-coding procedure with agreement statistics and the harmonization pass, pagination rule for the seven studies without printed page numbers). PRISMA-style flow diagram generated by corpus/make_prisma.py from the screening-record counts, with checksum self-tests. One anchor citation (Rojon et al. 2021, p. 196) read column-wise from the PDF and footnoted per convention; a candidate passage on p. 195 was identified as a Tranfield quotation inside Rojon and rejected as a secondary source. The three benchmark studies are cited as entities without page numbers (no content claim). Human-voice pass applied. Text reviewed and edited by the author before sending.